

Experimental Investigation of the Thermal Performance and Pressure Loss in Additively Manufactured mini-channels

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Abstract

Industrial gas turbines reach temperatures of 1500-2000K at high rotational velocities which means that much effort is spent on the design of an efficient cooling system. With the recent advances of the additive manufacturing (AM) industry, new design opportunities have open up for many industries and applications, including the design of cooling systems. However, a significant surface roughness will be present in AM components compared to traditionally manufactured components. An increased surface roughness inside a channel will affect both the heat transfer and pressure loss. The performance of AM channels are therefore not fully known and needs to be examined experimentally on the actual material to fully capture the effects of the increased surface roughness.

The aim with this project is to experimentally investigate the thermal performance and pressure losses experienced in AM channels due to surface roughness. This was done by using a Steady State Heat Transfer rig which was assembled and verified. AM and aluminium test channels were mounted in a copper block which was insulated and heated up by electrical heaters. The test channels were then subjected to an air flow of constant mass flow. Temperature and pressure measurements were made at the inlet and outlet together with mass flow measurements and copper block temperature measurements.

The Nusselt number and Darcy friction factor were used to evaluate the heat transfer and pressure losses experienced in the channels. The results showed that the heat transfer and friction factor increased significantly for the AM channels compared to smooth channels. Both the heat transfer and friction factor increased when the relative roughness of the channels increased.

This project was executed at Siemens Energy in Finspång at the Fluid Dynamic Laboratory and is a part of the work of obtaining thermal performance data for mini-channels manufactured by AM.

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Nomenclature

247	AM material CM247LC
939	AM material Inconel939
Δp	Pressure drop over channel [Pa]
Δp_{diff}	Pressure drop over channel with differential measurement [Pa]
$\dot{m}$	Mass flow [g/s]
ρ	Density [kg/m^3]
ε	Average sand grain diameter [m]
ε/D	Relative surface roughness parameter [-]
A_c	Cross section area of channel [m^2]
A_s	Surface area of channel [m^2]
AM	Additive manufacturing
c	Speed of sound [m/s]
c_1	Pressure loss coefficient at inlet
c_2	Pressure loss coefficient at outlet
C_p	Specific heat capacity [J/kgK]
D	Channel diameter [m]
D_h	Hydraulic diameter of channel [m]
f_d	Darcy friction factor [-]
h	Heat transfer coefficient [W/m^2K]
k	Conductivity coefficient [W/mK]
L	Length [m]
L/D	Ratio between length and diameter of channel [-]
M	Mach number [-]
Nu	Nusselt number [-]

Nu_G	Nusselt number for Gnielinski correlation [-]
P	Channel perimeter [m]
p	Pressure [Pa]
p_{atm}	Atmospheric pressure [Pa]
p_{dyn}	Dynamic pressure [Pa]
p_{inlet}	Pressure at inlet [Pa]
p_{outlet}	Pressure at outlet [Pa]
p_{stat}	Static pressure [Pa]
p_{tot}	Total pressure [Pa]
Pr	Prandtl number [-]
Q	Heat [W]
q	Heat flux [W/m^2]
R_a	Arithmetic Average Surface Roughness [m]
R_p	Maximum profile peak height [m]
R_t	Maximum height of the profile [m]
R_z	Average maximum height of the profile [m]
Re	Reynolds number [-]
T	Temperature [$^{\circ}C$] or [K]
T_s	Temperature of a surface [K]
T_{∞}	Temperature of free stream flow [K]
T_{cu}	Copper temperature [$^{\circ}C$] or [K]
T_{inlet}	Temperature at inlet [$^{\circ}C$] or [K]
T_{outlet}	Temperature at outlet [$^{\circ}C$] or [K]
T_w	Temperature of inner wall of test channel [$^{\circ}C$] or [K]
U	Velocity of flow [m/s]
$u()$	Uncertainty of measured input()
u_c	Combined uncertainty
U_{∞}	Velocity of free stream flow [m/s]

Introduction

1.1 Background

A gas turbine is an internal combustion engine which is used to convert thermal or chemical energy into mechanical energy. The principle is used for power generation applications and for aircraft propulsion. A simplified schematic of a gas turbine system can be seen in Figure 1.1 and the gas turbine SGT-800 manufactured by Siemens Energy can be seen in Figure 1.2.

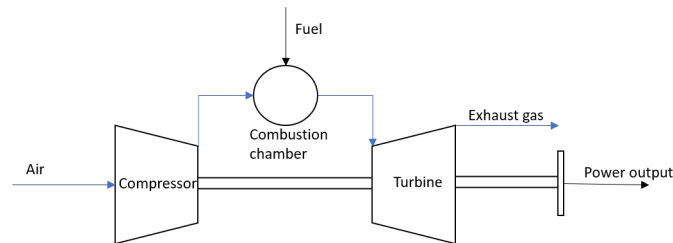


Figure 1.1: Simplified schematic of a gas turbine system [1]

One of the most important factors that will affect the performance of a gas turbine is the turbine inlet temperature. By increasing the inlet temperature, the work output of the gas turbine increases as well. Due to an increased inlet temperature, modern industrial gas turbines can be exposed to temperatures of 1500-2000K with high rotational velocities. No suitable materials can withstand such extreme temperatures and loads for a long period of time, therefore an efficient cooling system is necessary for a successful design of a gas turbine. Internal air cooling with forced convection is described to be one of the most successful methods for cooling systems in production engines. Components that are to be cooled does therefore contain several small cooling channels that can vary in design in order to achieve an effective surface area. It is desired that the channels achieve high heat transfer with low pressure losses. [1]

The recent advances in the additive manufacturing (AM) industry, also known as 3D printing, has open up new opportunities within many industries and applications, including the design of cooling systems for industrial gas turbines. The use of AM components gives more freedom in the design phase due to less geometrical constraints and lower the production time and cost compared to traditional manufacturing techniques. However, the performance of these new materials and components made from AM methods is not fully known and needs to be examined carefully before being used.

One difference between traditionally manufacturing and AM is that an increase in surface roughness is introduced for AM components. An increased surface roughness for cooling channels will affect

both the pressure loss and heat transfer of the internal flow. These effects need to be examined experimentally on the actual materials in order to fully capture the effects of the increased surface roughness.

Pressure losses and heat transfer in pipe or channel flow has been studied for several decades and the applications can be found in a number of industries. Nikuradse [2] performed experimental studies to examine the correlation between the pressure losses for pipe flow due to surface roughness and Reynolds number. Sand grains were applied to pipes and the roughness was characterized by using the average-sand-grain diameter compared to the diameter of the pipe. This translated to the relative roughness parameter (ε/D_h) which is still used in pressure loss measurements today.

Huang et al. [3] studied internal flow in channels with a diameter between 0.2-3 mm, also known as mini-channels, with a high relative roughness up to 42%. These dimensions are similar of the channels evaluated in the present study. They observed a deviation from the theoretical laminar value of $64/Re$ for the friction factor for the channels with a relative roughness above 7%. It was also reported that the transitional Reynolds number decreased with an increasing relative roughness. Similar conclusion was presented by Dai et al. [4] which observed that a roughness of less than 1% has little effect on the flow characteristics and suggests that $\varepsilon/D_h = 1\%$ could be used as a limit between rough and smooth mini channels.

Several studies about thermal performance in AM channels have been published from Pennsylvania State University (PennState). Stimpson et al. [5] performed experimental studies on channels manufactured by Direct Metal Laser Sintering (DMLS) using a cobalt-molybdenum-based superalloy (CrCo) powder with a relative sand grain surface roughness (ε/D_h) around 20-38% of the hydraulic diameter. They presented the results in the form of the Nusselt number and Darcy friction factor and showed that the friction factor increased significantly for the AM channels compared to smooth channels, especially for small hydraulic diameters of the channel. A significant increase in heat transfer was also presented, however, this increase was not proportional to the augmentation of the friction factor. The relative arithmetic mean roughness value of the surface roughness was presented in the form R_a/D_h , which seemed to agree well with the ε/D_h value.

This study examines test channels manufactured by the Selective Laser Melting (SLM) process with the low conductivity materials Inconel939 and CM247LC which are suitable for high temperature environments. The dimensions of the channels are of similar magnitude compared to cooling channels found in gas turbines while the shape of the channels may vary. The thermal performance of AM channels is highly dependent on the material and Inconel939 and CM247LC show great potential for being used in industrial gas turbines. Therefore, it is necessary to experimentally examine and evaluate their properties and performance before being implemented.

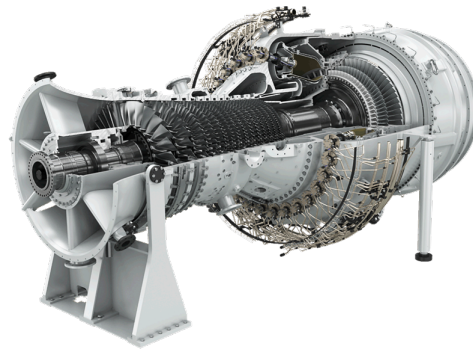


Figure 1.2: Gas turbine SGT-800, manufactured in Finspång, Sweden [6]

1.2 Study Objective

The aim of the project is to perform an experimental investigation of the thermal performance of AM channels for application in industrial gas turbines. A Steady State Heat Transfer (SSHT) test rig was used for investigating the heat transfer and pressure losses that occurs inside the AM channels due to surface roughness.

A test and measuring method, a data collection method and a post-processing method will be developed and the Nusselt number (Nu) and the Darcy friction factor (f_d) are the variables used to evaluate the heat transfer and pressure losses for the channels. The overall aim is to gather data of the performance for the AM channels which can be used in the design phase of the cooling system for gas turbines.

The test channels are mounted in a copper block which is insulated and heated up by electrical heaters. Copper is used because of its high conductivity which ensures a uniform temperature distribution around the test channel. The channel will then be subjected to an air flow at constant temperature and mass flow. Temperature and pressure measurements are made at the inlet and outlet of the test channel to calculate the Nusselt number and the Darcy friction factor.

An experimental method was chosen to investigate the thermal performance. The effects of the increased surface roughness can be captured fully by performing an experimental investigation on the actual AM materials.

1.3 Project Plan and Time Plan

The project took place at the Fluid Dynamic Laboratory at Siemens Energy in Finspång, Östergötland over the span of 8 months with start in August 2020. The project was divided into different phases with room for adjustments. A time plan is presented in Figure 1.3.

- Review of literature and background study
- Assemble and prepare test rig
- Perform Risk Analysis
- Validate test rig and test method by a bench marking process
- Insert changes to test rig and test method if necessary
- Run experiment for AM test channels of varying geometries
- Develop post-processing method for measured data evaluation
- Analyse and validate results
- Present results

Week	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Literatur review																																				
Assemble test rig																																				
Risk Analysis																																				
Benchmarking																																				
Changing of rig																																				
Run experiment																																				
Analyse and validate																																				
Write report																																				
Present results																																				

Figure 1.3: Overall schedule of project

1.4 Limitations

The project is an experimental project where focus is on the measured data evaluation. Fluid dynamic simulations were performed in addition to the experimental measurements for the use of validation and comparison of the measured results. These simulations are therefore not included as results.

Theoretical Background

2.1 Reynolds Number

One of the most important dimensionless quantities in fluid mechanics is the Reynolds number (Re) which is used to characterize the flow and its properties. The flow can be assumed to be laminar below certain limits of the Reynolds number and turbulent at higher Reynolds number with a transition zone in between. The Reynolds number for flow through a channel can be described as Equation 2.1 where the hydraulic diameter of the channel is the characteristic length. The hydraulic diameter is explained in Section 2.3.

$$Re = \frac{\rho U D_h}{\mu} = \frac{\dot{m} D_h}{\mu A_c} \quad (2.1)$$

D_h = hydraulic diameter [m]

A_c = cross sectional area of flow [m^2]

ρ = density of fluid [kg/m^3]

μ = dynamic viscosity of fluid [kg/ms]

U = velocity of fluid [m/s]

$\dot{m}$ = mass flow [kg/s]

Laminar flow in channels can be assumed when the streamlines of the flow are straight and parallel to each other and with a parabolic velocity distribution that does not fluctuate randomly with time for fully developed laminar flow. Many correlations and variables in fluid dynamics assumes fully developed flow, including the Nusselt number and Darcy friction factor, where both the velocity and temperature profile of the flow is fully developed and constant along the flow for internal flow with heat transfer [7].

Turbulent flow has non-ordered streamlines which can be described as chaotic and the fluid elements have fluctuating velocities while the velocity profile is flatter for channel flow. The entrance length for internal channel flow is much shorter for turbulent flow compared to its laminar counterpart. The entrance length can be described as the length in the channel before the flow becomes fully developed and where entrance effects are significant in magnitude and affects the flow. For channels with large L/D , ratio between length and diameter of channel where L denotes the total channel length and D the diameter, these entrance effects will become less significant.

The transition between laminar and turbulent flow does not happen immediately. The flow instead alters between the two flow types for a certain range of the Reynolds number before fully taking the form of one of the flow types. This range in Reynolds number is called the transition zone.

2.2 Boundary layer

Flow over a surface will have a boundary layer due to the interaction between the fluid elements and the solid surface. These boundary layers can be divided between a velocity boundary layer and a thermal boundary layer.

The velocity of the flow at the solid surface can be assumed to be zero. At a certain distance away from the solid surface, the velocity is close to the free-stream velocity U_∞ of the flow. This distance is known as the boundary layer thickness where the limit $U = 0.99U_\infty$ is used to separate the boundary layer region from the rest of the fluid flow.

Due to friction forces and shear stress, where shear stress ($\tau = F_f/A_s$) is described as the friction force (F_f) per unit area (A_s), between the solid surface and the fluid layer closest to it, the fluid will attempt to drag along the solid. Since these forces are dependent on the surface of the solid, an increased surface roughness will affect the shear stress acting between the fluid and solid which in turn will affect the friction forces.

For the thermal boundary layer, it is assumed that the flow closest to the surface has the same temperature as the solid surface, T_s . The temperature will then gradually change in the flow as the distance from the surface increases until it reaches the same temperature as the free-stream flow. The limit $T - T_s = 0.99(T_\infty - T_s)$ is used to separate the two regions where T_∞ is the temperature of the free stream flow.

2.3 Hydraulic Diameter

The hydraulic diameter of a channel is defined in Equation 2.2. It is commonly used in channel flow work since it is a common variable used for both symmetric and non-symmetric channels.

$$D_h = \frac{4A_c}{P} \quad (2.2)$$

D_h = hydraulic diameter [m]

A_c = cross sectional area of flow [m^2]

P = wetted perimeter of channel [m]

As can be derived from Equation 2.2, the hydraulic diameter for a circular channel is equal to the physical channel diameter. For non-circular channels, such as rectangular channels, the hydraulic diameter is used as a suitable substitute variable compared to other dimensions of the channel. The wetted surface perimeter is defined as the perimeter of the cross-section of the channel and can be described as the perimeter where the flow is in contact with the channel wall. Figure 2.1 show which dimensions are measured for a circular and a rectangular channels in order to obtain the hydraulic diameter.

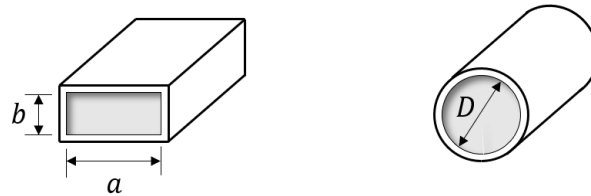


Figure 2.1: Rectangular and circular channel

2.4 Pressure losses

The pressure losses experienced in a channel flow are commonly presented using the non-dimensional variable Darcy friction factor, f_d , which quantifies the pressure losses due to the friction between the fluid and the channel walls. The Darcy friction factor is presented in Equation 2.3.

$$f_d = 2 \frac{D_h}{L} \frac{\Delta p}{\rho_{mean} U_{mean}^2} \quad (2.3)$$

D_h = hydraulic diameter [m]

L = length of channel [m]

Δp = pressure loss over channel [Pa]

ρ_{mean} = average density of fluid [kg/m^3]

U_{mean} = average velocity of fluid [m/s]

2.4.1 Friction pressure loss

The most significant pressure loss experienced in a channel flow is generally over the length of the channel, Δp , illustrated from point 2 to 3 in Figure 2.2 which show a similar setup compared to the experimental setup. This pressure loss will be affected of the increase of the surface roughness on the inside of the channel due to the increased friction between the fluid and the solid material. This pressure loss is also dependent on the length of the channel.

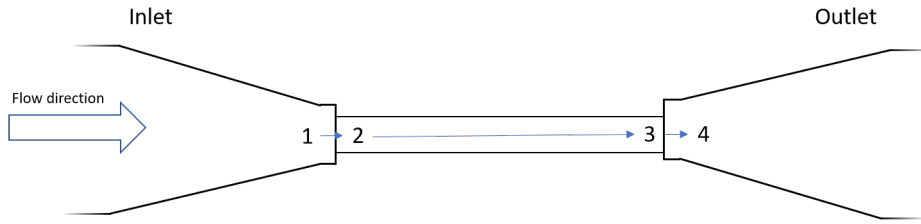


Figure 2.2: Pressure loss segments in channel

In the case of laminar flow, this pressure loss is mainly dependent on the Reynolds number for channels with low roughness. The theoretical value for f_d for laminar flow is $f_{d,laminar} = 64/Re$ which is valid for fully developed laminar flow. The Colebrook-White equation, Equation 2.4, describes the friction factor for fully developed turbulent flow dependent on the Reynolds number and relative roughness, ε/D_h [8].

$$\frac{1}{\sqrt{f_d}} = -2 \log_{10} \left(\frac{\varepsilon}{3.7 D_h} + \frac{2.51}{Re \sqrt{f_d}} \right) \quad (2.4)$$

The parameter relative roughness, ε/D_h is used to characterize the roughness of the wall and originates from Nikuradse [2] work from 1933, where ε is the average sand-grain roughness diameter. The Darcy friction factor together with the Colebrook-White equation and the laminar line are commonly presented in a Moody diagram [9] to illustrate how the friction factor is dependent on the Reynolds number and relative roughness of the channel.

2.4.2 Inlet and outlet pressure loss

Pressure losses are also experienced at the inlet and outlet of the channel due to the sudden contraction and expansion in the geometry, illustrated as point 1 to 2 and point 3 to 4 in Figure 2.2. These

pressure losses can be calculated with Equation 2.5 and 2.6. Idelčik [10] described correlations due to geometry for calculations of the coefficients c_1 and c_2 . For channels with 90° sharp edge at the inlet and outlet, the coefficients can be assumed to be $c_1 = 0.5$ and $c_2 = 1$ for turbulent flow and $c_1 = 0.5$ and $c_2 = 2$ for laminar flow. The difference of c_2 between laminar and turbulent flow is dependent on the difference of the velocity profiles between the two flow types.

$$p_{loss,in} = c_1 \frac{1}{2} \frac{\dot{m}^2}{\rho_i A_c^2} \quad (2.5)$$

$$p_{loss,out} = c_2 \frac{1}{2} \frac{\dot{m}^2}{\rho_o A_c^2} \quad (2.6)$$

$\dot{m}$ = mass flow [kg/s]

A_c = cross section area [m^2]

ρ = density at inlet and outlet [kg/ m^3]

As mentioned previously, the most significant pressure loss is generally over the length of the channel. However, depending on the L/D of the channel, the inlet and outlet pressure losses can become equally significant if they are in the same magnitude.

Minor pressure losses due to geometry are also experienced in other locations such as valves and connections but those are not included in this setup since they do not affect the pressure measurements.

2.4.3 Mach number

The Darcy friction factor is valid for incompressible fully developed flow. The flow can be assumed to be incompressible if the highest Mach number is below 0.3 [11]. The Mach number is a dimensionless variable that describes the ratio between the flow velocity and the speed of sound, Equation 2.7.

$$M = \frac{U}{c} \quad (2.7)$$

M = Mach number [–]

U = flow velocity [m/s]

c = speed of sound [m/s]

2.5 Heat Transfer

The heat transfer can be evaluated with the dimensionless variable called Nusselt number (Nu). It describes the ratio between convective and conductive heat transfer at a boundary, Equation 2.8, where the hydraulic diameter D_h is used as the characteristic length for internal channel flow.

$$Nu = \frac{h}{k/D_h} = \frac{hD_h}{k} \quad (2.8)$$

h = convective heat transfer coefficient [W/m^2K]

k = conductivity coefficient [W/mK]

D_h = hydraulic diameter [m]

Convective heat transfer is described as the transfer of heat from one region to another caused by the movement of fluid. There are two different types of convective heat transfer, natural convection

and forced convection. Forced convection happens when the fluid is forced over a boundary by for example a fan or a pressure tank. Internal and external convection is also used as a description where internal convection occurs when the fluid is enclosed by a solid boundary, for example for internal fluid flow in a channel.

Conductive heat transfer is described as the transfer of heat through a matter or material. The heat transfer occurs in the direction from high temperature to low temperature in the material. Conductive heat transfer can also be described as energy transfer from more energetic particles to less energetic particles due to the interaction between said particles.

Both conductive and convective heat transfer need matter or material to be able to transfer energy. The difference is that for convective heat transfer, the matter needs to have a bulk motion while this is not the case for conductive heat transfer.

For fully developed laminar flow in circular channels, the Nusselt number can be assumed to be constant and the surface roughness and friction factor effects are negligible. This constant value is $Nu = 3.66$ when a constant surface temperature is applied as boundary condition and $Nu = 4.36$ when a constant surface heat flux is applied as boundary condition[11].

For the case when fluid flows over a solid edge, conduction can be assumed from the edge to the fluid. Convection will then transport the heat in the direction of the fluid flow. Since the conduction occurs at the border between the fluid and solid, the surface of the solid will affect this heat transfer. An increased surface roughness changes the surfaces significantly compared to a smooth surface which therefore means that it will affect the heat transfer.

The convective heat transfer coefficient for flow in a channel is calculated by Equation 2.9.

$$h = \frac{Q}{A_s T_{LM}} \quad (2.9)$$

Q = heat input [W]

T_{LM} = logarithmic mean temperature [K]

A_s = surface area [m^2]

For internal flow with forced convection, the heat can be described with Equation 2.10 while the logarithmic mean temperature of the flow can be calculated with Equation 2.11

$$Q = \dot{m} C_p (T_o - T_i) \quad (2.10)$$

$$T_{LM} = \frac{T_o - T_i}{\ln((T_s - T_i)/(T_s - T_o))} \quad (2.11)$$

T_i = inlet temperature [K]

T_o = outlet temperature [K]

T_s = surface temperature [K]

C_p = specific heat capacity [J/kgK]

2.5.1 Correlations for smooth channels

Previous experimental studies have presented correlations for smooth channels for the Nusselt number, two of them being the Gnielinski correlation [12] and Dittus-Boelter correlation [7]. They can be expressed with Equation 2.12 and 2.13 respectively.

$$Nu_{Gnielinski} = \frac{(f/8)(Re - 1000)Pr}{1 + 12.7(f/8)^{1/2}(Pr^{2/3} - 1)} \quad (2.12)$$

$$Nu_{Dittus-Boelter} = 0.023Re^{4/5}Pr^n \quad (2.13)$$

Re = Reynolds number [-]

Pr = Prandtl number [-]

f = friction factor [-]

n = 0.3 for cooled flow and 0.4 for heated flow

The Prandtl number describes the ratio between momentum diffusivity and thermal diffusivity and is only dependent on the fluid and the fluid state. The Prandtl number is typically around 0.71 for air. [11]

The Gnielinski correlation is valid for fully developed turbulent flow for $3000 \leq Re \leq 5 \times 10^6$ and $0.5 \leq Pr \leq 2000$ while the Dittus-Boelter correlation is valid for $Re \geq 10000$ and $0.6 \leq Pr \leq 160$.

The augmentation of the heat transfer due to increased surface roughness is generally presented as Nu/Nu_0 over the range of Reynolds number where Nu_0 represent the Nusselt number for a smooth channel. The Gnielinski correlation will be used as Nu_0 in this study.

The Gnielinski correlation will also be used as a known correlation for an infinite long channel for the Nusselt number, Nu_∞ when evaluating the entrance effects in this study.

2.5.2 Entrance effects

It is important to evaluate the entrance effects when studying channel flow, especially for low L/D where entrance effects becomes increasingly significant. Mills [13] presented correlations for evaluating entrance effects for heat transfer in low L/D smooth channels. The entrance region heat transfer results were described with the dimensionless form Nu_x/Nu_∞ , where Nu_x is the Nusselt number at point x in the channel and Nu_∞ is the Nusselt number for fully developed conditions where the entrance effects are no longer present. In this study, the Gnielinski correlation has been used for Nu_∞ since it is valid for fully developed turbulent flow. The augmentation that is dependent on the L/D of the channel can be applied to Nu_∞ to compensate for the entrance effects. This augmentation for a channel with 90° sharp edge inlet is described in Equation 2.14.

$$\frac{Nu_x}{Nu_\infty} = 1 + \frac{8.7}{\frac{L}{D} + 5} \quad (2.14)$$

The entrance region can also be described as the entrance length, meaning the region of the flow before the velocity and temperature profile of the flow is fully developed.

2.6 Additive Manufacturing

Additive manufacturing is also known as 3D printing and is the name of the building process where a layer by layer method is used. Several different techniques and methods have been developed within AM and the techniques continues to develop at a rapid pace. The test channels used in this study are manufactured by a method called Laser Powder Bed Fusion method (LPBF), also known as Selective Laser Melting (SLM), and is a reliable method for metal AM components. The material is in the form of powder and is added one layer at a time in very thin layers onto a building platform. After every layer has been applied, a laser beam melts and fuses the powder together at the desired areas, forming a solid material. The platform then lowers and a new layer of material is added. This is then repeated and the melted material connects between the layers which will finally form a solid component. The process can also be seen in Figure 2.3.

There are many advantages of using additive manufacturing for example it is time efficient, cost efficient and less wasteful. One of the biggest advantages is the increased degree of freedom in designing by the removal of geometric constraints compared to using traditional manufacturing techniques.

The channels tested in this study were manufactured with EOS M290 machines, and the metal material powder underwent full melting.

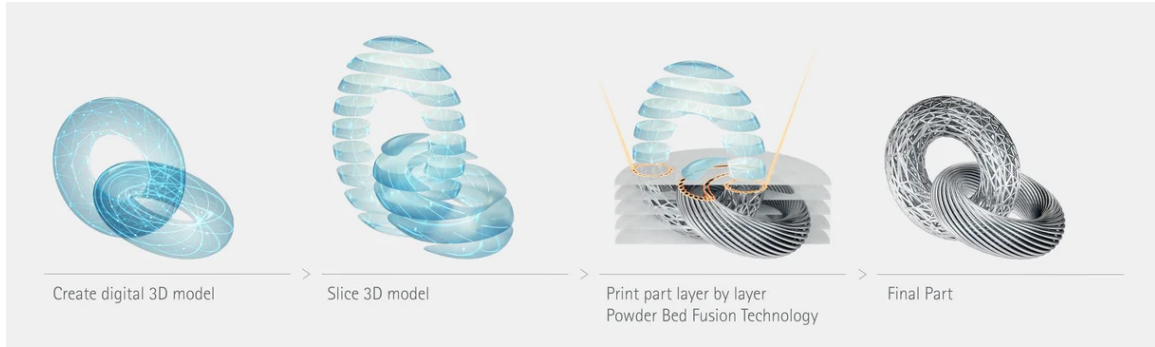


Figure 2.3: SLM manufacturing process. [14]

2.7 Uncertainty Analysis

Uncertainty analysis, or error analysis, is the process of estimating how great an effect the uncertainties in the individual measurements have on the calculated results. It is standard to perform an uncertainty analysis in an experimental project and to present the result of the analysis together with the main results. One method for error estimation is described by Kline and McClintock [15] where the combined uncertainty (u_c) of the input quantities (X_N) are calculated, Equation 2.16. It is first defined which input quantities (X_N) the variable of interest is depended on, Equation 2.15.

$$Y = f(X_1, X_2, \dots, X_N) \quad (2.15)$$

$$u_c^2(Y) = \sum_{i=1}^N \left(\frac{\partial f}{\partial X_i} \right)^2 u^2(X_i) \quad (2.16)$$

Method and Equipment

3.1 Experiment setup

The experimental setup is illustrated in Figure 3.1 and consist of a pressurized air system, a heater system and a test rig compartment. Test channels are located in the test rig compartment and accesses the pressurized air flow by turning on an external manual ball valve. The mass flow is then measured before entering the test rig compartment by a Coriolis MASSFLOW600 meter. Two different mass flow meters are used, one is used for mass flows between 0-1.1 g/s and the other is used for mass flows between 0.5-2 g/s . There are regulating valves on both the inlet and the outlet of the test compartment which is used to regulate the mass flow and pressure ratio between the inlet and outlet. The heaters are turned on manually by setting the power output. Four PT-100 thermocouples are attached to the experiment setup. One is attached at the inlet and one at the outlet for measuring the air temperature. The other two measures the temperature in the copper block at different radial distances from the test channel. There are 10 differential pressure measurements attached, five at the inlet and five at the outlet. They measure the static differential pressure relative to either the atmospheric pressure or the pressure at some other measurement point, for example between the inlet and outlet. The atmospheric pressure is measured by a Rosemount 3051C Absolute Pressure Meter. The measurements and their locations are listed in Table 3.1.

Table 3.1: Experimental measurements and their location

Measurement	Unit	Device	Location
Air temperature	°C	PT-100	Inlet/Outlet
Copper temperature	°C	PT-100	Copper block
Differential Pressure	Pa	PSI9116	Inlet/Outlet
Absolute Pressure	Pa	Rosemount	Atmosphere
Mass flow	g/s	Coriolis	Upstream of test compartment

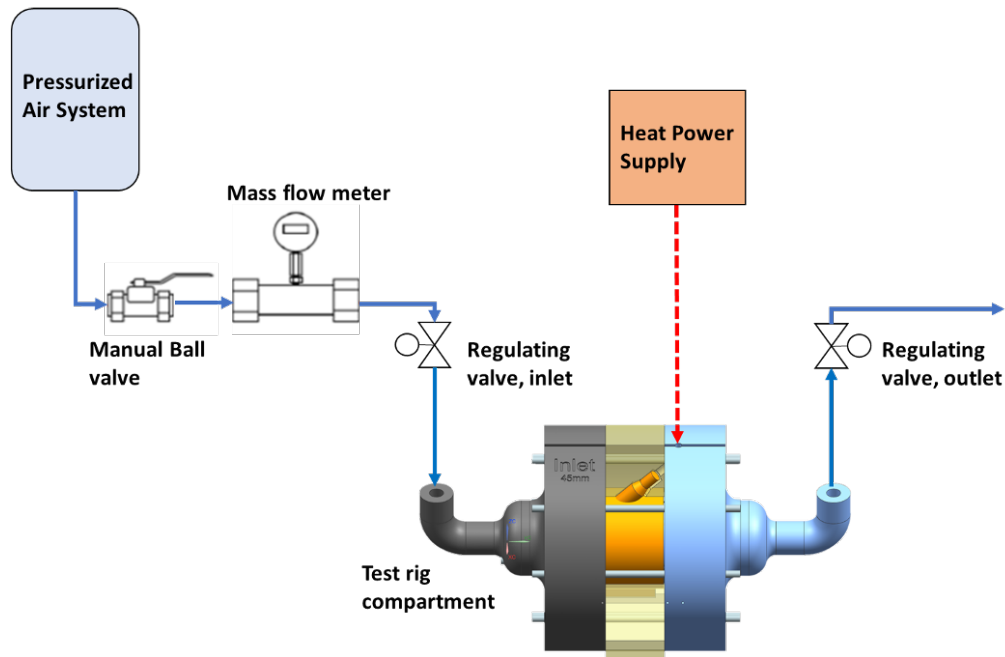
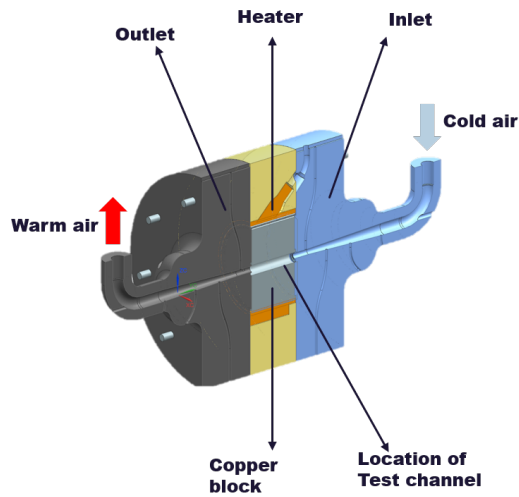
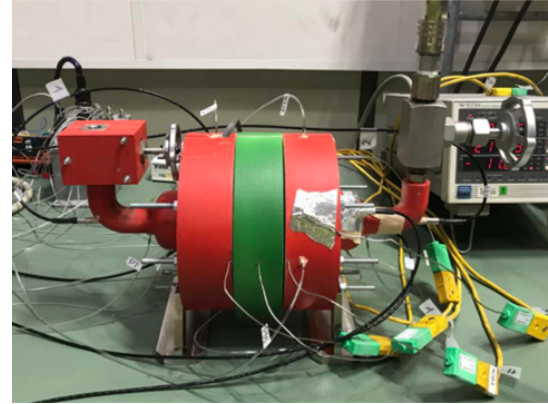


Figure 3.1: Schematics of experimental setup showing the pressurized air system, heat power supply and test rig compartment

3.1.1 Test rig compartment

The main components of the experimental test rig compartment are the inlet, outlet, copper block and heaters as illustrated in Figure 3.2. Surrounding the copper block and heaters are also plastic insulation parts. The test channels are placed in the copper block with a thin layer of thermal paste between the channel and copper block. The thermal paste is also applied between the copper block and heaters. Thermal paste is used to ensure as much contact as possible between the components. The length of the test rig compartment refers to the length of the test channel adapted for the test rig, namely $L=45$ mm.

The test channel is located in a copper block due to the high conductivity of the copper, which will yield a uniform temperature distribution in the copper block and around the outer wall of the test channel. A photo of the test rig compartment can also be seen in Figure 3.2.

(a) Model of test compartment $L=45\text{mm}$ (b) Photo of test compartment $L=45\text{mm}$ Figure 3.2: Test compartment for $L=45\text{mm}$

3.1.2 Heater system

The heaters are so called nozzle elements, Figure 3.3, and are located around the copper block. Two heaters with different widths are used in the setup. They are connected to an external power source where the power is manually set.



Figure 3.3: Example of nozzle element heaters [16]

3.1.3 Measurement devices

A schematic of the measurement devices can be seen in Figure 3.4. The atmospheric pressure and mass flow measurement are connected to a DataScan7220 which is connected to the PC via USB. The differential pressure, PT-100 temperature and mass flow measurements are connected to the PC via Ethernet cable. Additional information about the devices are presented in Table 3.2 and in Appendix B.

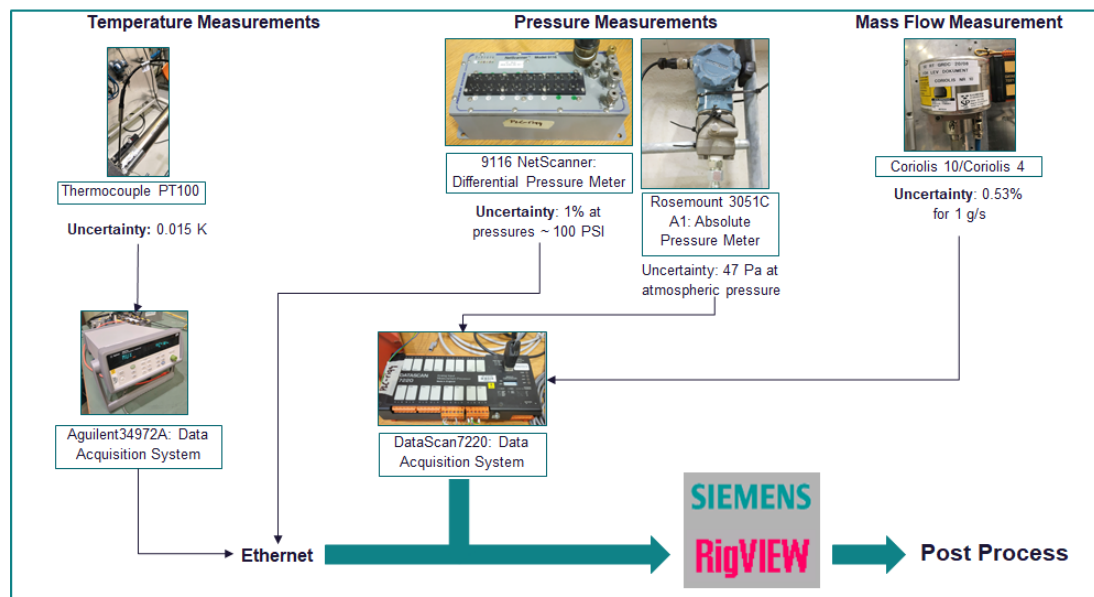


Figure 3.4: Measurement instrumentation setup and connection to PC

Table 3.2: Measurement devices used in experiment setup

Device	Type	Unit	Amount	Uncertainty	Range
Coriolis 10	Mass flow	g/s	1	0.53%	0-1.1 g/s
Coriolis 4	Mass flow	g/s	1	0.53%	0-2 g/s
Rosemount 3051C	Absolute Pressure	Pa	1	46 Pa	-
9116 NetScanner	Differential Pressure	Pa	10	1%	0-100 PSI
PT-100	Temperature	°C	4	0.015K	0-250 °C

3.2 Test channels

A total of 6 Nickel alloy AM test channels were tested. They consisted of two different materials, Inconel939 and CM247LC of length 45mm. All of the test channels in this study are classified as mini-channels when using the same classification as Kandlikar et al. [17], see Table 3.3. One of the test channels had a rectangular shape while the rest were circular i.e axisymmetric. Two aluminium channels were also tested as a part of the validation process of the test rig since they could be assumed to be smooth channels manufactured by traditional methods.

Table 3.3: Channel classification where D is the minimum channel dimension [17]

Type of channel	Channel size
Conventional channels	$> 3 \text{ mm}$
Minichannels	$3 \text{ mm} \geq D > 200 \mu\text{m}$
Microchannels	$200 \mu\text{m} \geq D > 10 \mu\text{m}$
Transistional microchannels	$10 \mu\text{m} \geq D > 1 \mu\text{m}$
Transistional nanochannels	$1 \mu\text{m} \geq D > 0.1 \mu\text{m}$
Nanochannels	$0.1 \mu\text{m} \geq D$

Microscopic measurements of the test channels were made since the design and printed dimensions may not be identical with the present state of additive manufacturing. The measured dimensions from the microscopic measurements were used in the calculations. Photos of the inlet of test channels with hydraulic design diameter 0.75mm and 1.5mm can be seen in Figure 3.5, for both AM material 247 and 939 and also one aluminium channel. The microscopic measurements can also be seen in Figure 3.5.

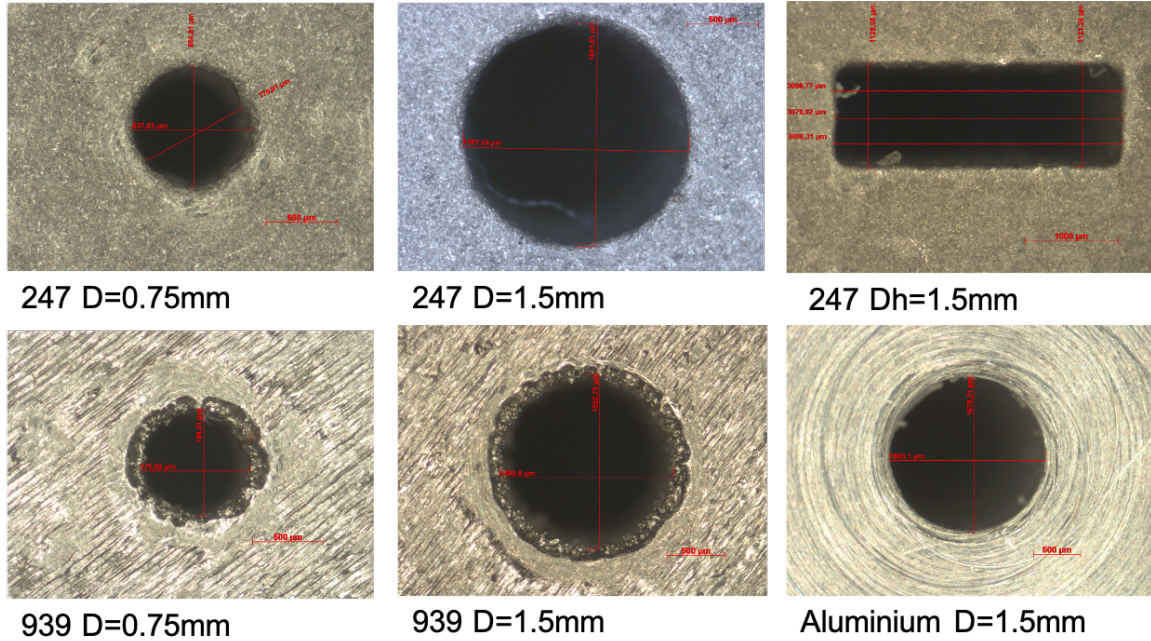


Figure 3.5: Photo of test channels for $L=45\text{mm}$ from microscopic measurements

All test channels with design and measured dimensions are presented in Table 3.4 for the circular

channels while the rectangular channel is presented in Table 3.5. The variables a and b in Table 3.5 refers to the length of the sides of the rectangular channel shape.

Table 3.4: Design and measured dimensions for circular test channels in [mm]

Material	D_h (design)	D_h (measured)	L/D (design)	Length
247	0.75	0.7625	60	45
939	0.75	0.7672	60	45
Aluminium	1.5	1.67	30	45
247	1.5	1.583	30	45
939	1.5	1.517	30	45
247	3	3.132	15	45
Aluminium	3	2.928	15	45

Table 3.5: Design and measured dimensions of rectangular test channel in material 247 in [mm]

D_h (design)	a (design)	b (design)	a (measured)	b (measured)	D_h (measured)
1.5	3	1	3.089	1.128	1.652

3.2.1 Surface roughness

It is expected that material 939 has a greater surface roughness compared to material 247. Surface roughness measurements were made of five of the test channels. Four of the resulting surface roughness profiles can be seen in Figure 3.6-3.9 where two surface profiles are from 247 channels and two are from 939 channels. Surface profiles shows the height of the peaks and valleys of a surface sample of a material. From the profile, several surface roughness variables can be extracted.

By comparing the profiles in Figure 3.6 and 3.7, it is evident that the 939 channel has a higher magnitude in height, which supports the statement that the 939 material has a higher surface roughness than 247. The arithmetic mean roughness value, R_a , was extracted from roughness measurements. R_a is the average roughness between a roughness profile and the mean line and is often used to measure and categorize a surface roughness. This value is higher for the 939 channel than the 247 channel of similar hydraulic diameter, namely $R_a = 0.0143\text{mm}$ and $R_a = 0.0071\text{mm}$ respectively.

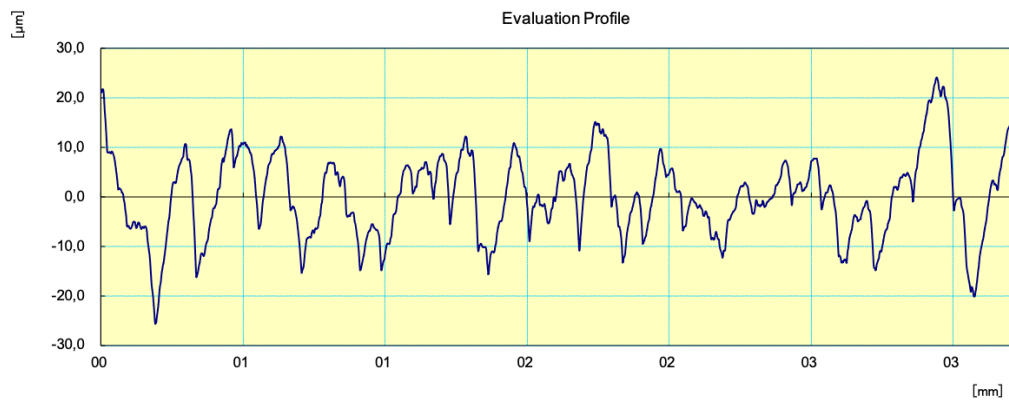


Figure 3.6: Surface roughness profile for test channel 247 $D_h=1.5\text{mm}$ rectangular. $R_a = 0.0071\text{mm}$

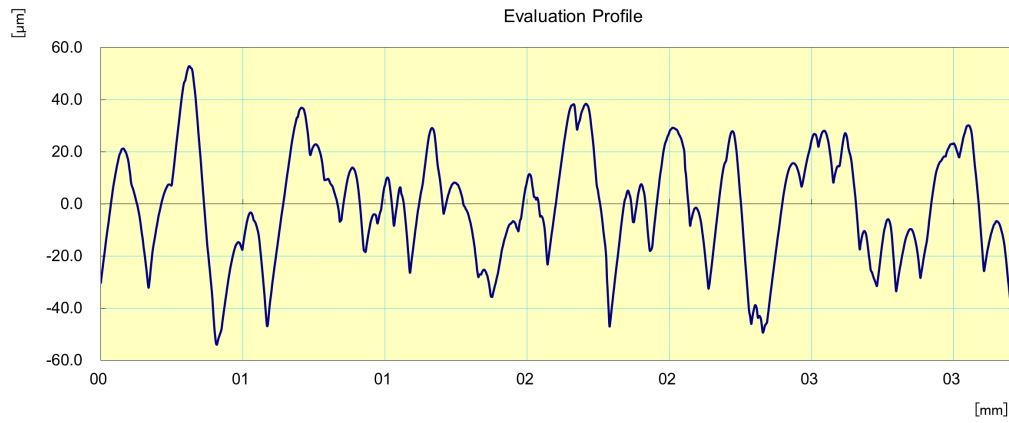


Figure 3.7: Surface roughness profile for test channel 939 $D_h=1.5\text{mm}$. $R_a = 0.0142\text{mm}$

However, even if the surface roughness is expected to be greater for material 939 compared to 247, this is not true for all cases and the surface roughness differs between channels of the same material as well. By comparing the two test channels with $D_h = 0.75\text{mm}$ in Figure 3.8 and 3.9, the 247 channel shows a higher surface roughness than the 939 channel. The R_a value is also higher for the 247 channel, namely $R_a=0.0163\text{mm}$ for the 247 channel while it is $R_a=0.0142\text{mm}$ for the 939 channel.

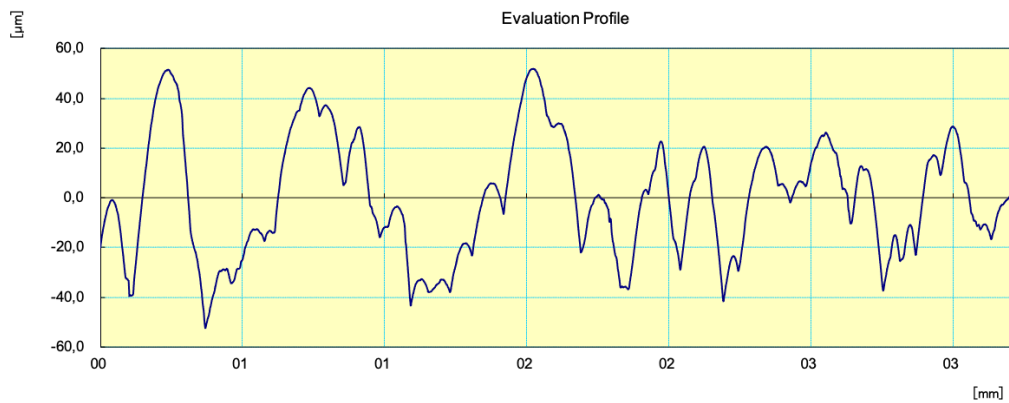


Figure 3.8: Surface roughness profile for test channel 247 $D_h=0.75\text{mm}$. $R_a = 0.0163\text{ mm}$

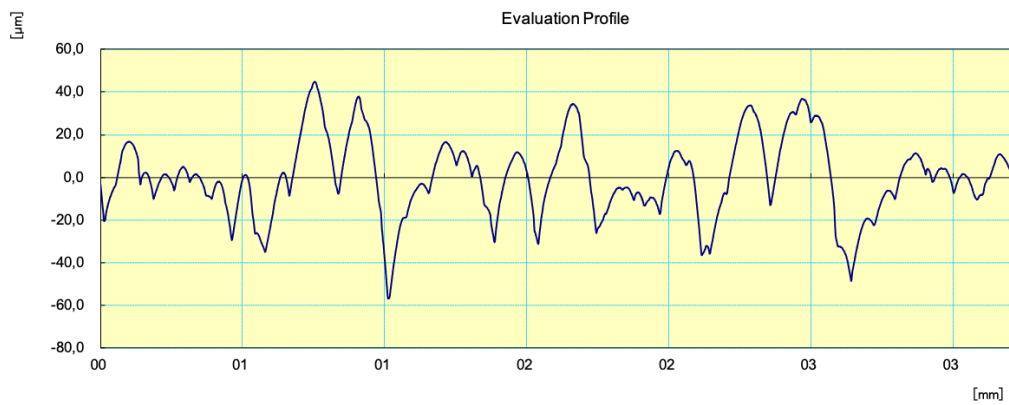


Figure 3.9: Surface roughness profile for test channel 939 $D_h=0.75\text{mm}$. $R_a = 0.0142\text{mm}$

More information about the surface roughness measurements and their results are described in Section 4.2.1.

3.3 Test Procedure

Two types of tests were performed, so called cold test and hot test, where the main difference is the use of heaters. Both hot tests and cold tests were performed and analysed for every test channel. The friction factor was evaluated for the cold test and the Nusselt number was evaluated for the hot test.

The first step in both test procedures is to turn on the external air flow. The regulating valves are then used to achieve the desired mass flow and pressure ratio between the inlet and outlet. Subsequently, the heaters are turned on for the hot test and data collecting is started. The rig needs to heat up for approximately two hours before changing to the next operating point i.e a different

mass flow. Thereafter, the mass flow is changed approximately once every hour since this is the time needed for the system to stabilize before gathering data.

The cold test is very similar to the hot test except that no heaters are used. Therefore, the experiment does not need to heat up and mass flows can be changed approximately every 5-10 minutes.

The test procedures for the two tests methods are also presented in Table 3.6 and 3.7.

It is desirable to avoid compressible effects since the friction factor is not valid for compressible flow. The Mach number is therefore evaluated for all operating points. If the Mach number reaches above 0.3, the test is redone and the inlet and outlet pressures are adjusted. By increasing the pressure at the outlet, the pressure in the entire system increases and a higher Mach number can be avoided by getting a pressure ratio between outlet and inlet that is much lower. The Mach number is evaluated at the outlet since this is where the flow velocity is expected to be the highest due to the sudden expansion from the test channel into the plastic outlet.

Table 3.6: Test procedure for hot test

Step	Action
1	Check that all tubes and cables are connected properly
2	Check that all measurements devices are ON
3	Turn on RigView program on PC
4	Create text file where data will be stored
5	Enable system in RigView and check Channel monitor
6	Turn ON heaters to desired effect
7	Wait until copper temperature reaches 50-70 °C
8	Turn ON external ball valve to pressurised air system
9	Start writing to text file in RigView
10	Regulate valves at inlet and outlet to reach desired pressure ratio and mass flow
11	Observe data and wait until Nusselt number stabilizes
12	Log data over 10 seconds in RigView and store to a second text file
13	Repeat step 10-12 for all operating points
14	Stop writing to text file
15	Turn OFF heaters
16	Wait until copper temperature goes below 50 °C
17	Turn OFF external ball valve to pressurised air system
18	Disable system in RigView

Table 3.7: Test procedure for cold test

Step	Action
1	Check that all tubes and cables are connected properly
2	Check that all measurements devices are ON
3	Turn on RigView program on PC
4	Create text file where data will be stored
5	Enable system in RigView and check Channel monitor
6	Turn on RigView CLIENT program and monitor pressure channels
7	Turn ON external ball valve to pressurised air system
8	Start writing to text file in RigView
9	Regulate valves at inlet and outlet to reach desired pressure ratio and mass flow
10	Observe pressures in RigView CLIENT and wait until they stabilizes
11	Log data over 10 seconds in RigView and store to a second text file
12	Repeat step 9-11 for all operating points
13	Stop writing to text file
14	Turn OFF external ball valve to pressurised air system
15	Monitor pressures and wait until experiment depressurizes
16	Disable system in RigView

3.4 Data Sampling

The software used to read and collect data was the in-house program *Siemens RigView v3.3, Gas Turbine Measurement and Evaluation system*. The trend program *RigViewClient* was used to observe data in real time and to help determine when pressures were stabilized. Data was sampled in two different ways, once every second and an averaged data sample over 10 seconds for every operating point. For every test, two different text files with data was therefore created. The main data file used in the calculations was the one with the averaged data.

3.5 Post-processing

The methods used to obtain the variables f_d and Nu are summarized in Table 3.8. Both the experimental and theoretical methods are included since the experimental results are compared with previously known correlations. More details about the experimental methods are described in Sections 3.5.2 and 3.5.3.

Table 3.8: Methods and equations used to obtain experimental and theoretically values

Variable	Flow region	Experimental	Theoretically
f_d	Laminar	$f_d = 2 \frac{D_h}{L} \frac{\Delta p}{\rho_{mean} U_{mean}^2}$	$64/Re$
	Turbulent		White-Colebrook (Equation 2.4)
Nu	Laminar	Nu method	$Nu = 3.46$
	Turbulent		Gnielinski (Equation 2.12)

3.5.1 Software

The data was post-processed in *Matlab R2019b* and *Python*. The in-house turbine cooling program *C3D v.3.1.23* for 3D and 1D simulations was also used. The Matlab and Python scripts created for the post-processing can be found in Appendix A.

3.5.2 Darcy Friction Factor

The Darcy friction factor was evaluated for every test channel. The density and velocity of the flow were evaluated at the inlet and outlet with Equation 3.1 and 3.2.

$$\rho = \frac{pM_{air}}{TR} \quad (3.1)$$

$$U = \frac{\dot{m}}{\rho A_c} \quad (3.2)$$

M_{air} = molar mass [kg/mol]

R = universal gas constant [J/molK]

The measured pressure in this set up is the static pressure of the flow. The Darcy friction factor is valid for the total pressure which is equal to the sum of the static pressure and the dynamic pressure, $p_{tot} = p_{stat} + p_{dyn}$. The dynamic pressure at the inlet and outlet was therefore evaluated as well, Equation 3.3.

$$p_{dyn} = \frac{1}{2}\rho U^2 \quad (3.3)$$

Pressure losses that arise at the inlet and outlet of the test channel due to the sudden contraction and expansions in geometry are presented in Chapter 2. These pressure losses were evaluated according to Equation 2.5 and 2.6. The coefficients c_1 and c_2 were used according to Table 3.9, where c_1 is the pressure loss coefficient for the inlet pressure loss and c_2 is for the outlet.

Table 3.9: Pressure loss coefficients for inlet and outlet

	c_1 (Inlet)	c_2 (Outlet)
Laminar flow	0.5	2
Turbulent flow	0.5	1

Two different methods were used to decide the pressure drop over the channel, Δp_{diff} and Δp . The first method uses a direct differential pressure measurement between the inlet and outlet called p_{diff} . The second method measure the pressure at the inlet and outlet separately by two different absolute pressure measurements. The evaluation of Δp for the two different methods are presented in Equation 3.4 and 3.5.

$$\Delta p_{diff} = p_{diff} - p_{loss,in} - p_{loss,out} + p_{dyn,in} - p_{dyn,out} \quad (3.4)$$

$$\Delta p = p_{tot,in} - p_{loss,in} - p_{loss,out} - p_{tot,out} \quad (3.5)$$

The Darcy friction factor is then evaluated according to Equation 2.3. The Matlab code for calculation of the Darcy friction factor can be seen in Appendix A.

3.5.3 Nusselt number

It is assumed that the inner wall temperature of the AM channels is not constant along the length of the channel since the conductivity of the AM materials are significantly lower than for highly conductive materials such as copper. An iterative method was therefore developed to evaluate the Nusselt number in the AM channels.

The temperature distribution in the test channel is assumed to be axisymmetric which is valid for the circular test objects. Discretization is made along the flow direction of the channel into 100 cross sections and local values of the air and wall temperatures are calculated. The outlet air temperature is obtained and compared with the measured value of the outlet temperature. This is repeated for different heat transfer coefficients (h) as input until the difference between the calculated and measured outlet temperature is less than 0.1K. The resulting average wall temperature and the iterated value of h is then used to calculate the Nusselt number. The Matlab and Python code for this method are located in Appendix B and a flow chart of the method is shown in Figure 3.12.

After dividing the channel into 100 cross sections, local values of the heat flux and wall temperature are calculated together with the guessed heat transfer coefficient. To calculate the local values of the heat flux, a 1D thermal model through the channel wall is used, Equation 3.6. This heat flux is then used to calculate the local value of the wall temperature, Equation 3.7.

$$q = \frac{T_{gas} - T_{cu}}{\frac{1}{2P_i h} + \frac{\ln(r_o/r_i)}{2\pi k} + \frac{1}{2P_o h_{contact}}} \quad (3.6)$$

$$T_w = T_{gas} - \frac{q}{P_i h} \quad (3.7)$$

q = heat flux per meter pipe [W/m]
 T_{gas} = local value of air temperature [K]
 T_{cu} = copper temperature [K]
 T_w = wall temperature [K]
 h = iterated heat transfer coefficient [W/m²K]
 P_i = inner perimeter of channel [m]
 P_o = outer perimeter of channel [m]
 r_i = inner radius of channel [m]
 r_o = outer radius of channel [m]
 k = conductivity coefficient of material [W/mK]

For non-axisymmetric objects, such as a rectangular channel, the in-house simulation program C3D was used with a similar method as for axisymmetric channels. A 3D model of the test channel was imported in C3D and boundary conditions were applied. The boundary conditions were the measured air inlet temperature, inlet and outlet pressure and copper temperature. By using different heat transfer coefficients (h) as input, the C3D model is iterated until mass flow and outlet air temperature are similar to the measured mass flow and outlet temperature. This method can of course be applied to symmetric channels as well.

A cut through model of a circular test channel in C3D can be seen in Figure 3.10. The large temperature gradient in the material is due to the low conductivity of the AM materials.

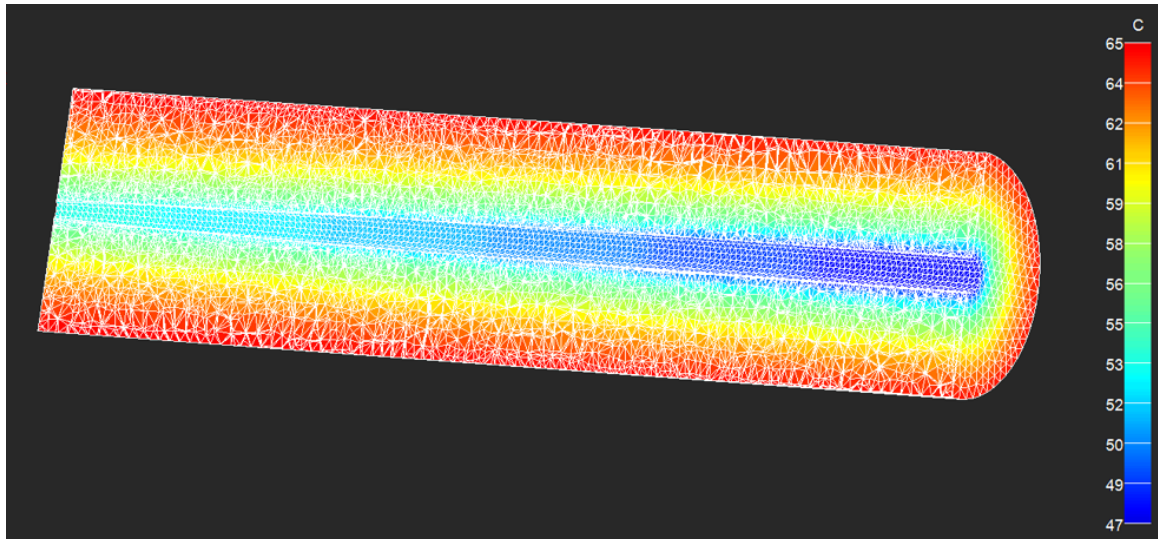


Figure 3.10: Model in C3D that shows temperature profile for a circular test channel in AM material

The assumption that the wall temperature is not constant along the channel was evaluated with these methods. The resulting temperatures for one operating point in the circular test channel 939 $D_h=1.5\text{mm}$ can be seen in Figure 3.11 using the axisymmetric method in Matlab. The inner wall temperature is not constant along the channel and there is a great temperature difference between the inner and outer wall of the AM test channel.

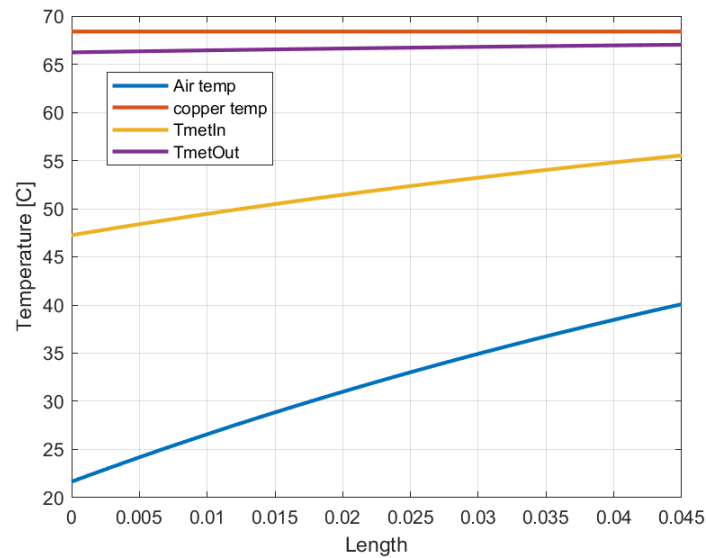


Figure 3.11: Temperatures over length in test channel 939 $D_h=1.5\text{mm}$ $L=45\text{mm}$. TmetIn = inner wall temperature. TmetOut = outer wall temperature

getOutletTemperature

Flow chart that describes method for obtaining calculated outlet temperature

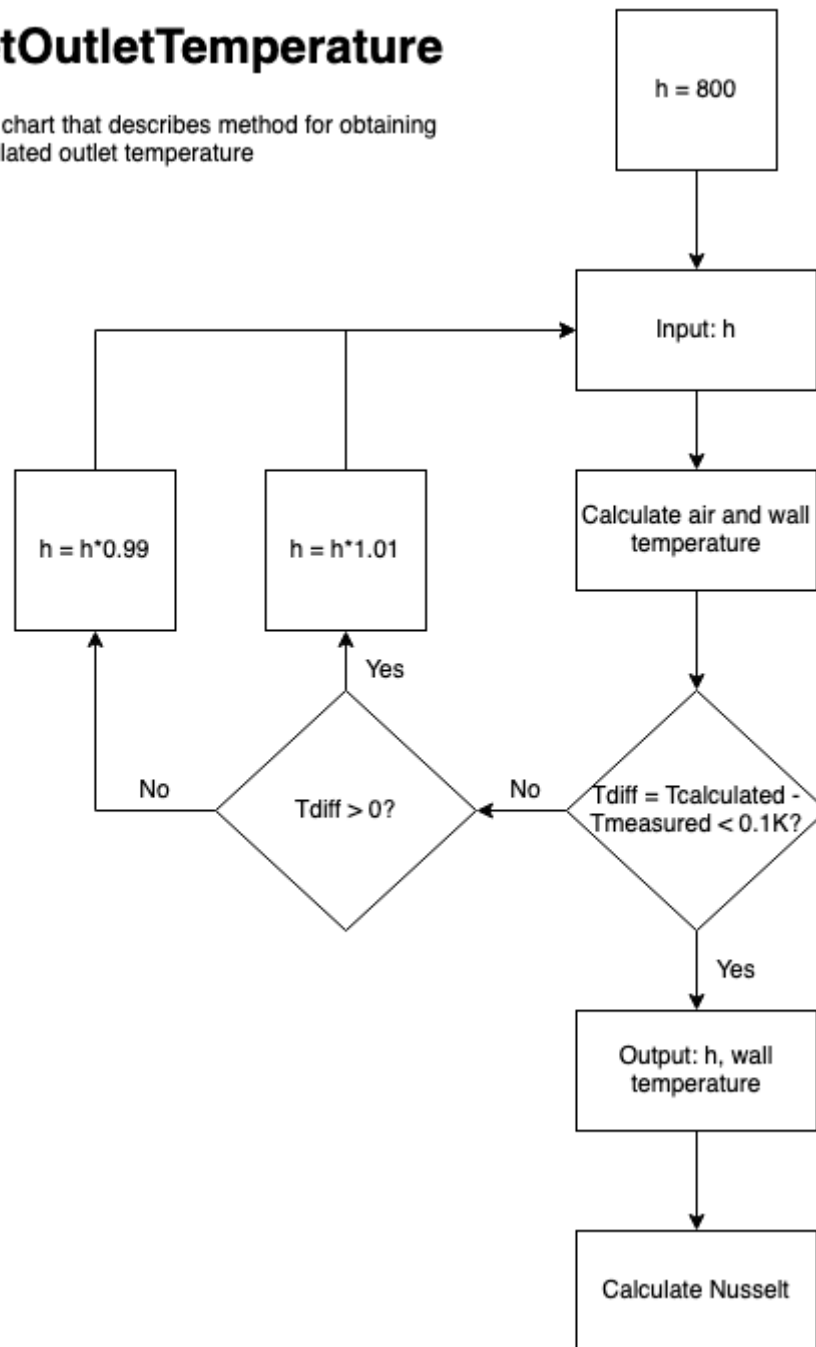


Figure 3.12: Flow chart of outlet temperature calculation method for Nusselt number evaluation

3.6 Uncertainty Analysis

An uncertainty analysis was performed using the method described in Chapter 2. The measured input variables for Darcy Friction factor was identified, Equation 3.8.

$$f_d = f(p_{diff}, p_{in}, p_{out}, p_{atm}, T_{in}, T_{out}, \dot{m}, D_h) \quad (3.8)$$

The Darcy friction factor is evaluated using Equation 2.3, the uncertainty expression is therefore identified in Equation 3.9.

$$u(f_d) = \sqrt{\left(\frac{\partial f_d}{\partial \Delta p} u(\Delta p)\right)^2 + \left(\frac{\partial f_d}{\partial D_h} u(D_h)\right)^2 + \left(\frac{\partial f_d}{\partial \rho_{mean}} u(\rho_{mean})\right)^2 + \left(\frac{\partial f_d}{\partial U_{mean}} u(U_{mean})\right)^2} \quad (3.9)$$

The uncertainties for the input quantities were obtained from data sheets for the measurement devices and are presented in Table 3.10. Some relevant data sheets can be found in Appendix B.

Similar method was used for the Nusselt number uncertainty analysis where the measured input variables are defined in Equation 3.10 and the uncertainty expression is presented in Equation 3.11.

$$Nu = f(T_{in}, T_{out}, T_{cu}, \dot{m}, D_h, k_{247/939}) \quad (3.10)$$

$$u(Nu) = \sqrt{\left(\frac{\partial Nu}{\partial h} u(h)\right)^2 + \left(\frac{\partial Nu}{\partial D_h} u(D_h)\right)^2 + \left(\frac{\partial Nu}{\partial k_{air}} u(k_{air})\right)^2} \quad (3.11)$$

Table 3.10: Uncertainties of measured input quantities

$u(T_{in})$	$u(T_{out})$	$u(D_h)$	$u(\dot{m})$	$u(p_{in})$	$u(p_{out})$	$u(p_{diff})$	$u(p_{atm})$	$u(k_{247})$	$u(k_{939})$
0.015 K	0.015 K	10 μm	0.53%	1%	1%	1%	46 Pa	0.21	0.89

Results

4.1 Bench Marking

Bench marking is the process used to validate an experimental setup. Circular aluminium test channels were used for the bench marking tests in this study. They can be assumed to be smooth channels manufactured by traditional methods. Therefore, the results from these channels can be compared with known correlations for smooth channels.

The results from two aluminium channels for the Nusselt number are presented in Figure 4.1a and 4.1b.

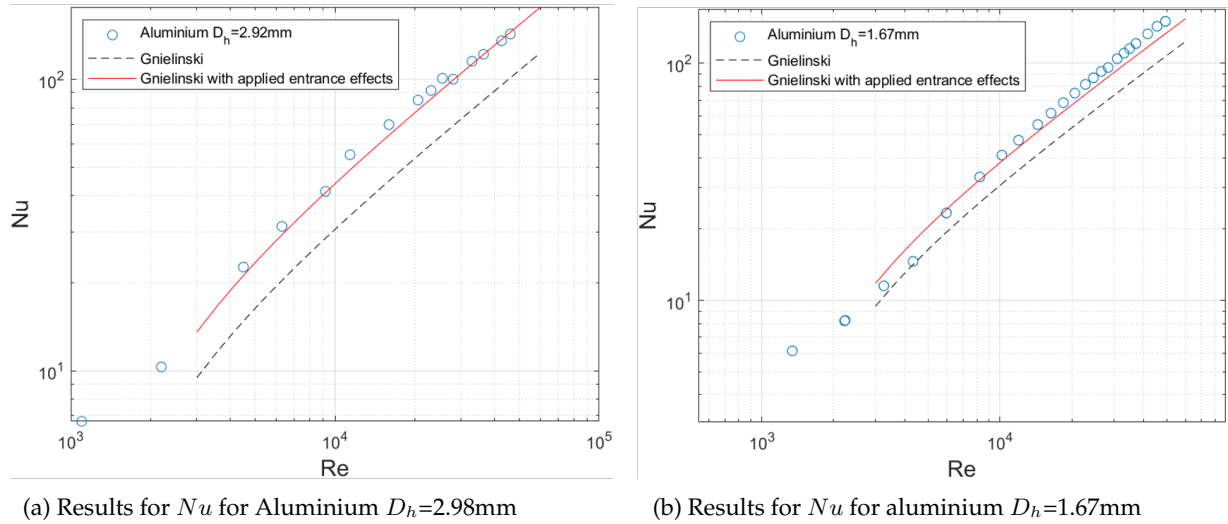


Figure 4.1: Bench marking results of Nusselt number for aluminium channels

The Gnielinski correlation is used for comparison for the Nusselt number. There is a deviation between the measured results and the Gnielinski correlation for the two presented aluminium channels. Therefore, entrance effects were evaluated. As described in Chapter 2, an augmentation can be applied to the Gnielinski correlation to compensate for entrance effects for low L/D channels. This has been done for these channels, which gives results that agrees well with the measured data for the Nusselt number for both aluminium channels

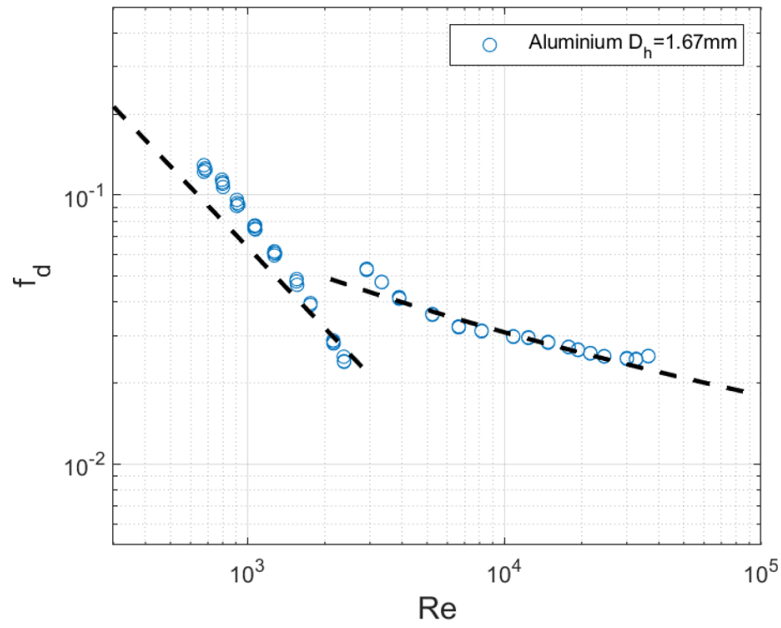


Figure 4.2: Results for f_d for aluminium $D_h=1.67$ mm

The results for the Darcy friction factor for the aluminium test channel with $D_h=1.67$ mm are presented in Figure 4.2 together with the correlation for laminar flow ($64/Re$) and the Colebrook-White equation for smooth channels where $\varepsilon = 0$ for turbulent flow. The friction factor results follow both the laminar and turbulent correlations.

These results indicate that the experimental set up is valid for Darcy friction factor and Nusselt number evaluations.

4.2 Darcy Friction Factor

The results for the Darcy friction factor are presented in Figure 4.3 for five AM test channels, including the rectangular one, together with one aluminium test channel. The results are plotted together with the laminar line ($64/Re$) and the Colebrook-White equation for $\varepsilon = 0$. The relevant equations for the Darcy friction factor results are presented in Section 2.4.

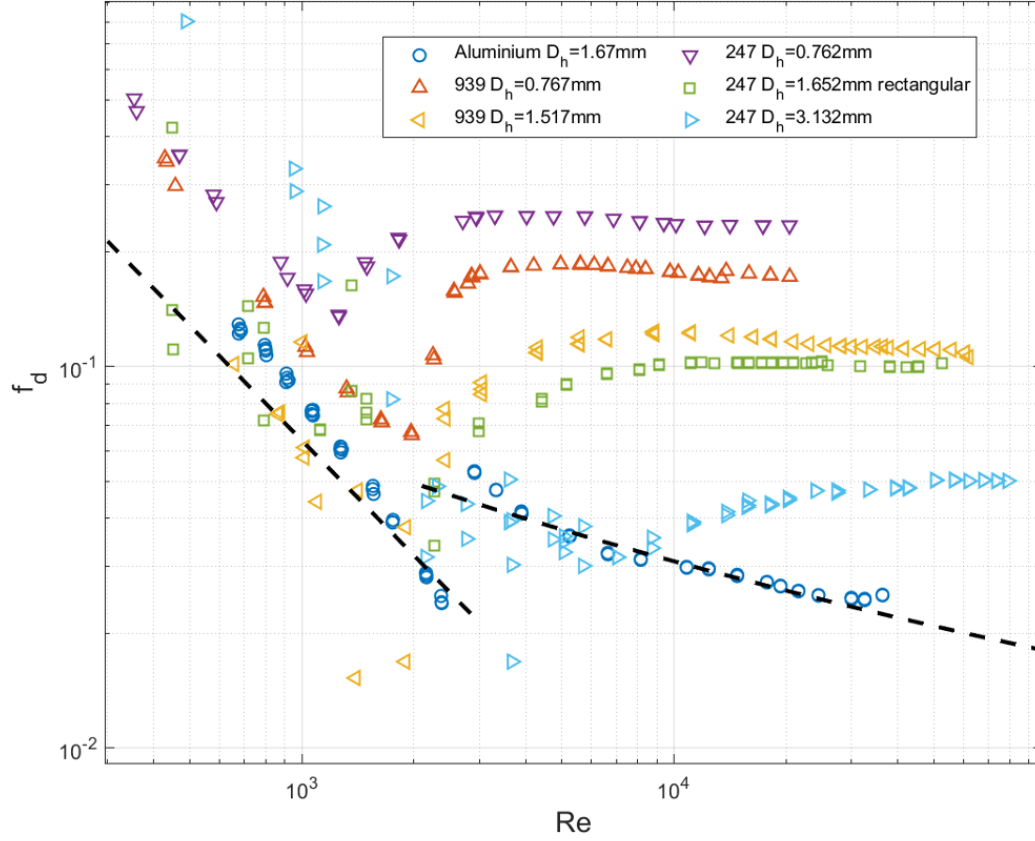


Figure 4.3: Results for Darcy friction factor for five AM channels

For lower Reynolds number, the friction factors for the five AM channel have similar behavior as the laminar line although a deviation from the line is present. This can be due to wrong estimations of the hydraulic diameter. However, a deviation from the laminar line has been observed in similar experimental studies, such as the study of Huang et al. [3], and can be due to a high relative roughness of the channels. Much of the previous experimental work used to develop the friction factor correlations used channels with a relative roughness of up to 5%, meaning that for channels with higher surface roughness, these correlations might not be the best fit.

For higher Reynolds, the friction factor for all AM channels converge towards constant values. This value increases with the relative roughness of the channel.

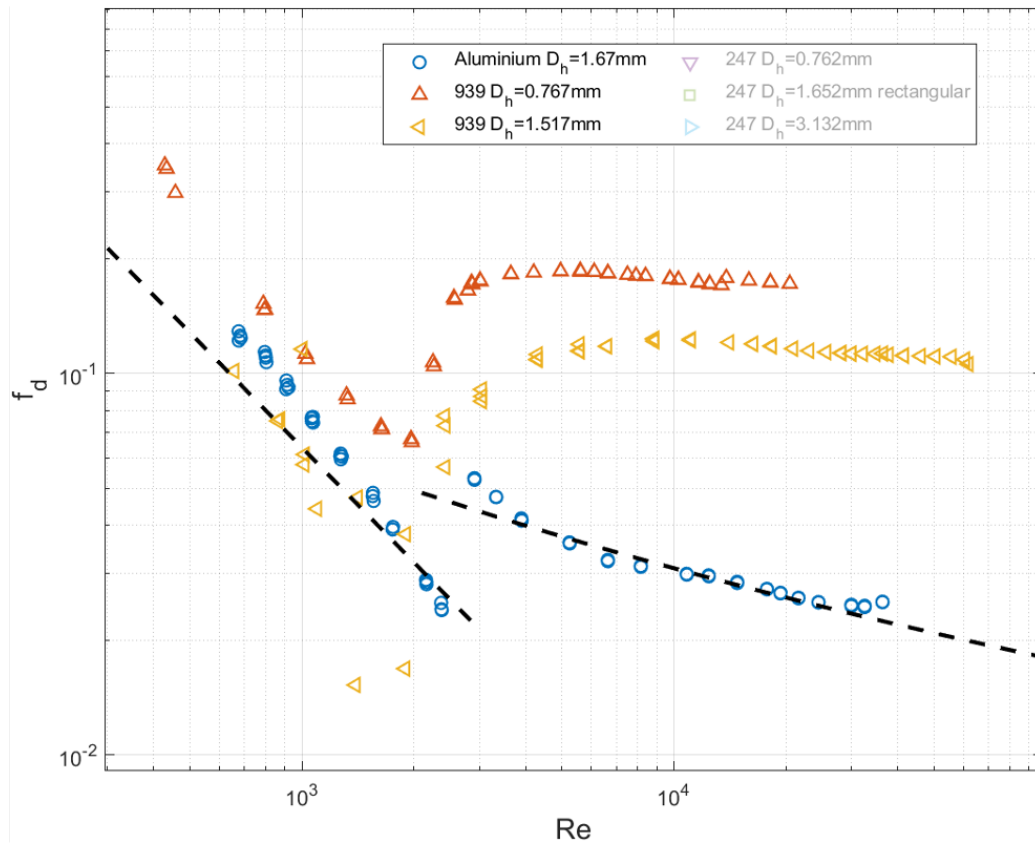


Figure 4.4: Friction factor results for 939 test channels and aluminium channel

By looking closer at the results for the 939 channels, Figure 4.4, it is evident that the friction factor is higher for the channel with the smaller hydraulic diameter for the same material. The friction factor for $D_h=0.75$ mm channel converges towards a higher constant value for higher Reynolds than that of the $D_h=1.5$ mm channel. Since the relative roughness describes the ratio between the sand grain diameter and channel diameter, this result is expected.

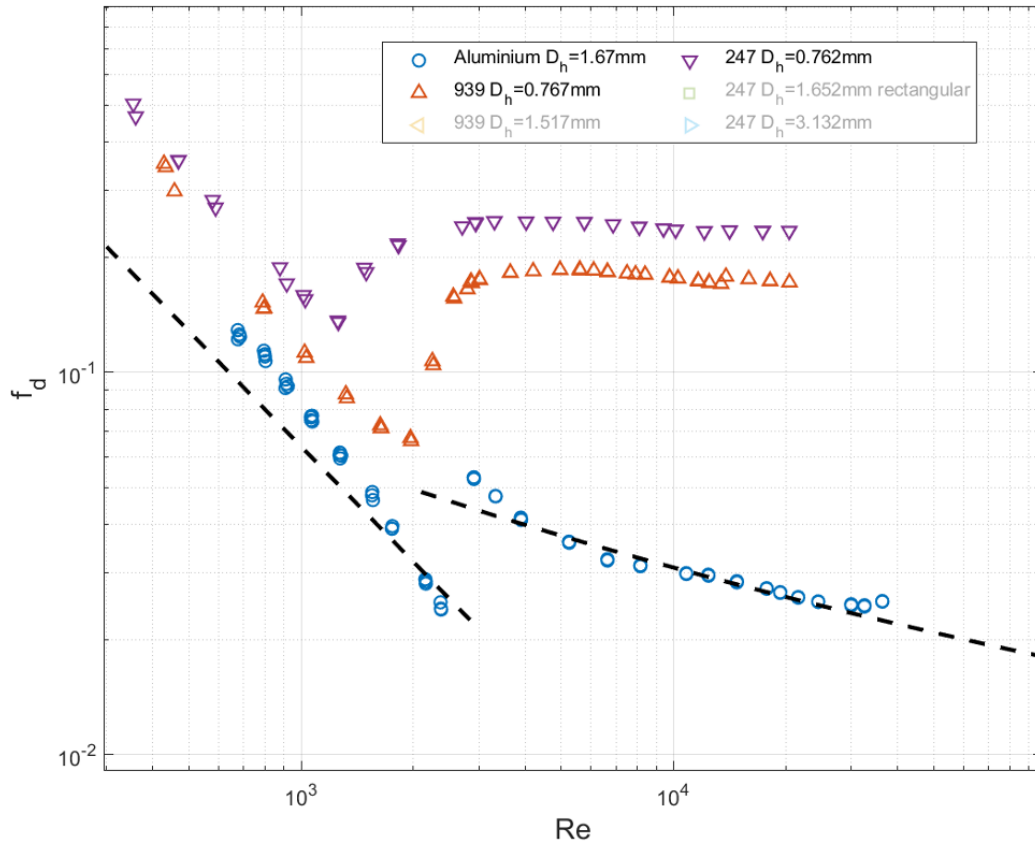


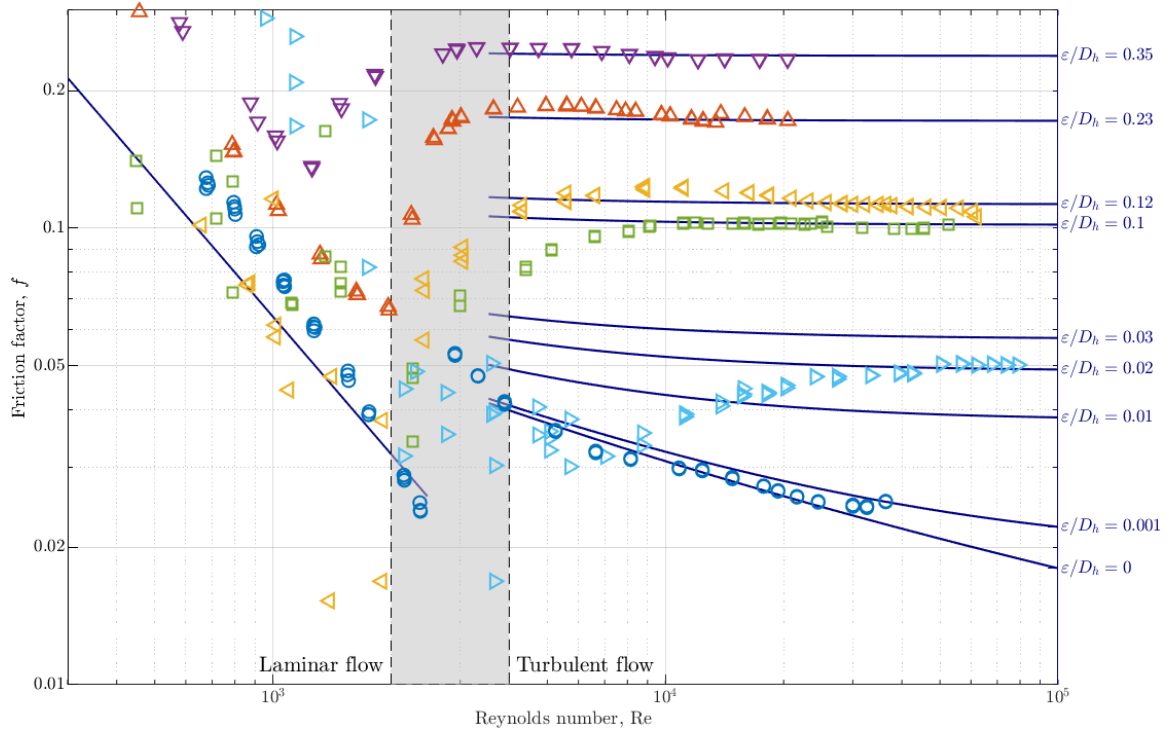
Figure 4.5: Friction factor results for $D_h=0.75\text{mm}$ test channels and aluminium channel

In Figure 4.5, the results for the two channels with diameter $D_h=0.75\text{mm}$ is presented. It is expected that the 939 channel would yield a higher friction factor and an earlier transition from laminar to turbulent flow than the 247 channel for approximately the same diameter, however, this is not the case. This may be due to a number of reasons, for example, that the hydraulic diameter is estimated incorrectly. Microscopic measurements of the diameter is only made at the inlet and outlet of the channel while the dimensions on the inside of the channel are still unknown, which leaves room for error in the estimation of the hydraulic diameter.

However, roughness measurements showed that the highest relative roughness belongs to the 247 channel which means that it would be accurate that it yields the highest friction factor. The results from the roughness measurements are discussed more in Section 4.2.1.

4.2.1 Surface roughness parameters

The estimation of the surface roughness can be described by the relative roughness variable, ε/D_h . Different values of ε/D_h are plotted with the Colebrook-White equation in Figure 4.6 together with the experimental results for the friction factor. The experimental results align with certain values for ε/D_h . From this, the sand grain roughness value ε can be estimated for the test channels, see Table 4.1.

Figure 4.6: Friction factor results together with ε/D_h valuesTable 4.1: Estimated ε values for test channels

Material	D_h [mm]	ε/D_h [-]	ε [mm]
939	0.7672	0.23	0.176456
939	1.517	0.12	0.18204
247	0.7625	0.35	0.266875
247	1.652	0.1	0.1652
247	3.132	0.02	0.06264

The ε value should be approximately the same for channels with the same material. This is the case for the two channels with material 939 where the values are 0.176 and 0.182. However, ε differs significantly for the channels with material 247.

Measurements of the surface roughness were made of the test channels as described in Chapter 3 where surface roughness profiles of the channels are presented. The resulting roughness parameters are presented in Table 4.2.

- R_a , the roughness average, describes the arithmetic average of the absolute values of the profile heights.
- R_p , maximum profile peak height, the distance between the highest point of the profile and the mean line.
- R_{tj} , Maximum height of the profile, the vertical distance between the highest and lowest point of the profile

- R_z , Average Maximum height of the profile, the average of the successive values of R_t calculated over evaluation length.

Table 4.2: Measured roughness values for test channels

Material	D_h [mm]	R_a [mm]	R_z [mm]	R_p [mm]	R_t [mm]
939	0.7672	0.0142	0.0809	0.0381	0.0999
939	1.517	0.0143	0.0699	0.0333	0.0885
247	0.7625	0.0163	0.0660	0.0341	0.0816
247	1.652	0.0071	0.0403	0.0192	0.0538
247	3.132	0.0080	0.0412	0.0238	0.0664

For the 939 channels, the roughness parameters are very similar to each other. This is also the case for the two 247 channels with hydraulic diameter of 1.652mm and 3.132mm. However, the 247 channel with diameter of 0.7625 stands out and have roughness values much higher than expected for the 247 material. This 247 channel has roughness values that are more similar to the 939 material.

Table 4.3: Measured roughness values expressed as relative roughness

Material	D_h [mm]	R_a/D_h [-]	R_z/D_h [-]	R_p/D_h [-]	R_t/D_h [-]
939	0.7672	0.0185	0.1055	0.0497	0.1303
939	1.517	0.0094	0.0461	0.0219	0.0583
247	0.7625	0.0214	0.0866	0.0447	0.1070
247	1.652	0.0043	0.0244	0.0116	0.0325
247	3.132	0.0025	0.0131	0.0076	0.0212

In Table 4.3, the measured roughness parameters are divided by the hydraulic diameters which creates relative roughness parameters. For the R_a/D_h parameter, the highest relative roughness is given by the 247 $D_h=0.75$ mm channel. This was also the case when describing the relative roughness in the form ε/D_h and it is also this channel that yields the highest friction factor. This shows that the roughness can differ between channels of the same material as well.

4.2.2 Uncertainty for friction factor

The uncertainty was evaluated according to the method described in Chapter 3 and the results for test channel 247 $D_h=0.75\text{mm}$ are shown in Table 4.4. The error results are approximately the same for all test channels.

Table 4.4: Uncertainty of Darcy friction factor for 247 D=0.75mm channel

Max error	Min error	Mean error
5.91 %	5.59 %	5.70 %

Each input quantity contributes to the combined uncertainty u_c , it was therefore evaluated which input contributed the most for a low and a high Reynolds numbers. The results are shown in Figure 4.7 where the contributions between the different input quantities are divided into percentages.

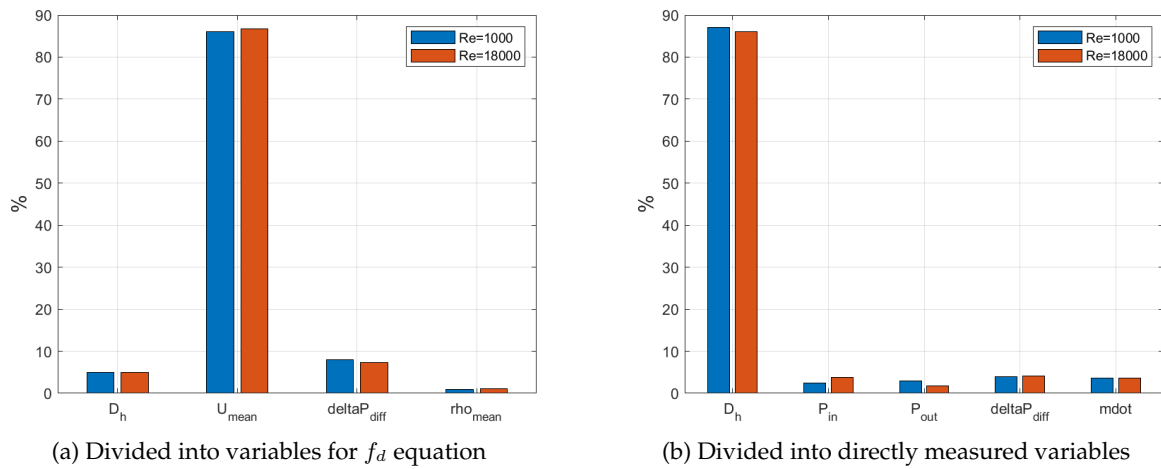


Figure 4.7: Uncertainty contribution for Darcy friction factor

In Figure 4.7a, the uncertainty contributions are divided between the different variables that are included in the Darcy friction factor equation, Equation 2.3. The velocity is the parameter that is contributing the most to the uncertainty.

In Figure 4.7b, the uncertainty contributions are instead divided into the variables that are directly measured in the experiment. The hydraulic diameter contributes to the uncertainty for approximately 85% for both high and low Reynolds numbers.

4.3 Nusselt number

The results for the Nusselt number are presented in Figure 4.8 and in the form Nu/Nu_0 in Figure 4.9 for six AM test channels, including the rectangular one, together with one aluminium test channel.

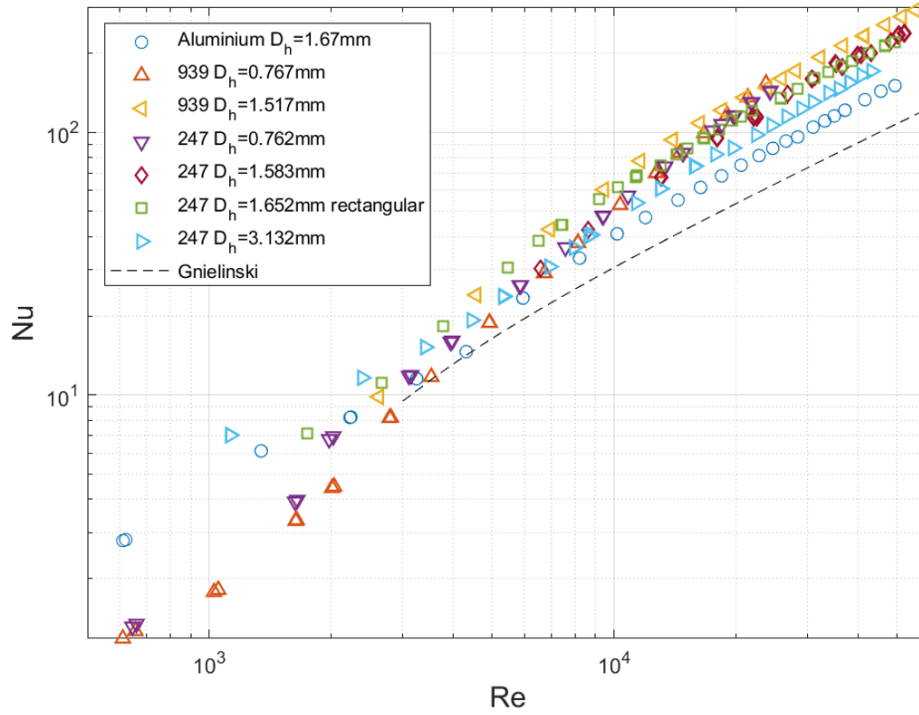


Figure 4.8: Results for the Nusselt number

The Nusselt number results are presented between approximately $600 < Re < 50000$. The Nusselt number should obtain a constant value for laminar flow or Reynolds below approximately $Re=2000$ [7]. However, this is not the case for the presented results. This can be due to an incorrect temperature measurement or assumption in the setup.

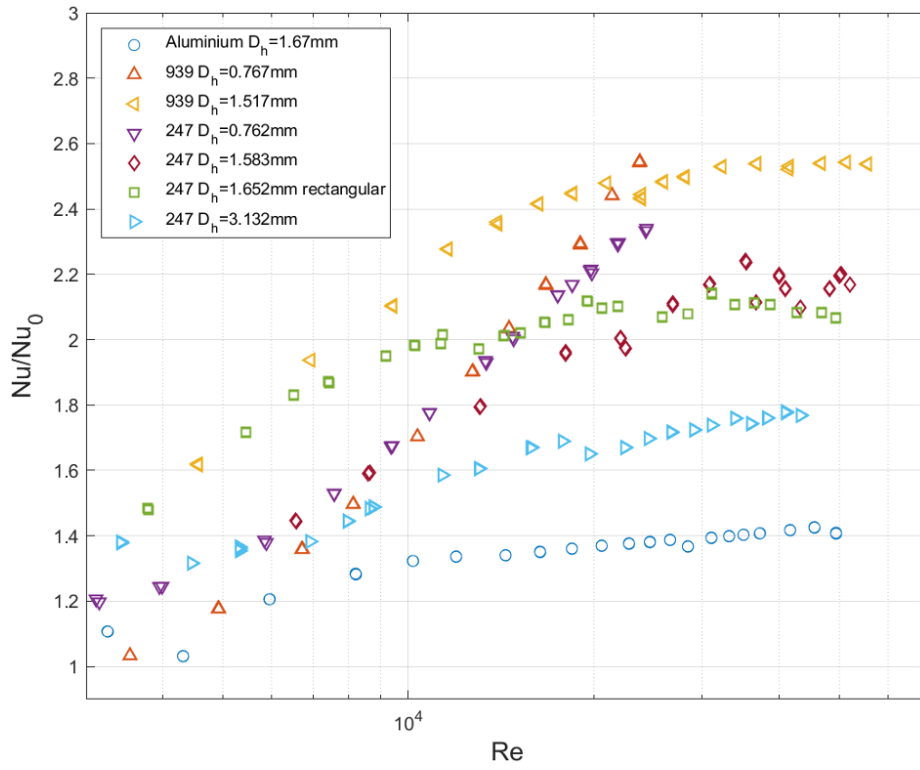


Figure 4.9: Results for Nusselt number in the form Nu/Nu_0 where the Gnielinski correlation is used as Nu_0

The increase of the Nusselt number approaches a constant value for higher Reynolds number although this is not the case for the channels with the smallest diameter, $D_h=0.75\text{mm}$, Figure 4.9. Higher Reynolds numbers could not be reached for these channels with the pressures available in this experiment setup. Therefore, it is not known if the Nu/Nu_0 would obtain a constant value for higher Reynolds for these two channels. However, the behaviour for these two channels are very similar and no major difference between the two materials is observed.

The highest increase in heat transfer is found in the 939 channel with $D_h = 1.5\text{mm}$, where the Nu/Nu_0 value stabilizes around 2.6 for higher Reynolds. Higher increase of the Nusselt number is expected for 939 due to the higher surface roughness.

Both a rectangular and circular channel with a hydraulic design diameter of 1.5mm in material 247 were tested for the Nusselt number. The difference between these two channels is relatively small for higher Reynolds numbers, indicating that the geometry is not significant for the heat transfer for this material. The Nu/Nu_0 value is around 2.2 for higher Reynolds which is a significant increase compared to smooth channels.

An increase of approximately 1.4 for the aluminium channel is also present and is most likely due to the cause of entrance effects. However, it should be noted that even if the aluminium channel is considered smooth compared to the AM channels, a relatively small surface roughness can still be present in this channel.

4.3.1 Uncertainty for Nusselt number

The uncertainty of the Nusselt number results was evaluated with the methods described in Chapter 3. The results for the uncertainty for channel 939 $D_h = 1.5\text{mm}$ are presented in Table 4.5. It is expected that the uncertainty are similar for all the channels.

Table 4.5: Uncertainty of Nusselt number for 939 $D_h=1.5\text{mm}$ channel

Max error	Min error	Mean error
7.93 %	2.71 %	4.85 %

Evaluation of which input contributed the most for a low and a high Reynolds numbers was made. The results are shown in Figure 4.10 where the contributions between the different input quantities are divided into percentages.

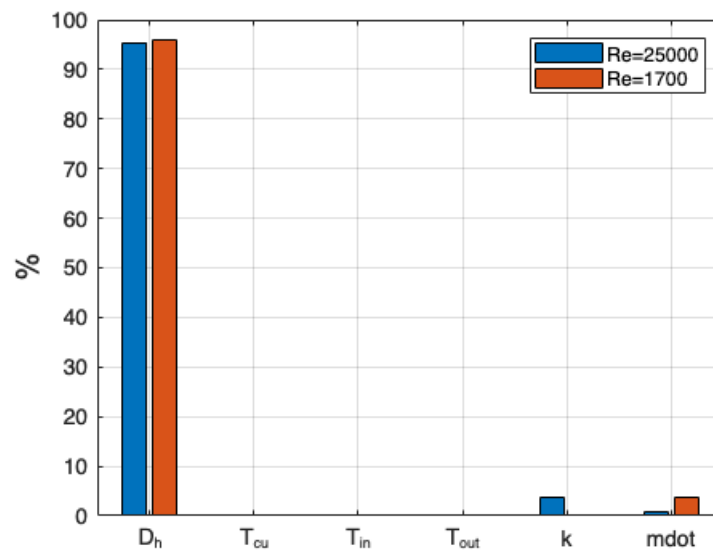


Figure 4.10: Uncertainty contribution to the Nusselt number

Once again, the hydraulic diameter is the variable contributing the most to the uncertainty.

4.4 Comparison with PennState

Similar studies about the thermal performance of AM channels for high temperature applications within gas turbine and aerospace technology have been performed at Pennsylvania State University where the results were presented as the Nusselt number and Darcy friction factor. They also presented measured R_a and R_a/D_h values of their channels [5]. Since the PennState study focus on the performance of AM channels due to surface roughness it is of great interest to compare the results between this project and their study.

The relative roughness using the R_a/D_h form is used to compare the channels between this study and the PennState study since values for R_a are given for both.

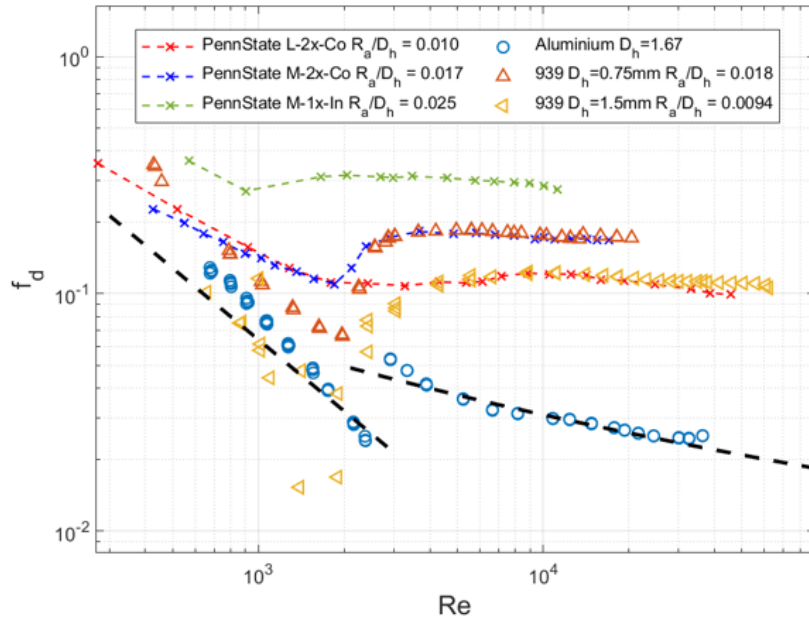


Figure 4.11: 939 channels comparison with PennState channels of similar relative roughness R_a/D_h

In Figure 4.11, the friction factor results for the two 939 channels are presented together with friction factor results from PennState with similar relative roughness, R_a/D_h . The 939 channel with hydraulic diameter of 0.75mm has a relative roughness of 0.018 and shows a similar trend and results of the friction factor as the AM channel with relative roughness of 0.017 from PennState. The same observation is made for the 939 channel with a relative roughness of 0.0094 that shows similar results as the channel with relative roughness of 0.010 from PennState. This is indicating that R_a/D_h could be a potential parameter of estimating the friction factor for AM channels since it yields similar results between two different studies.

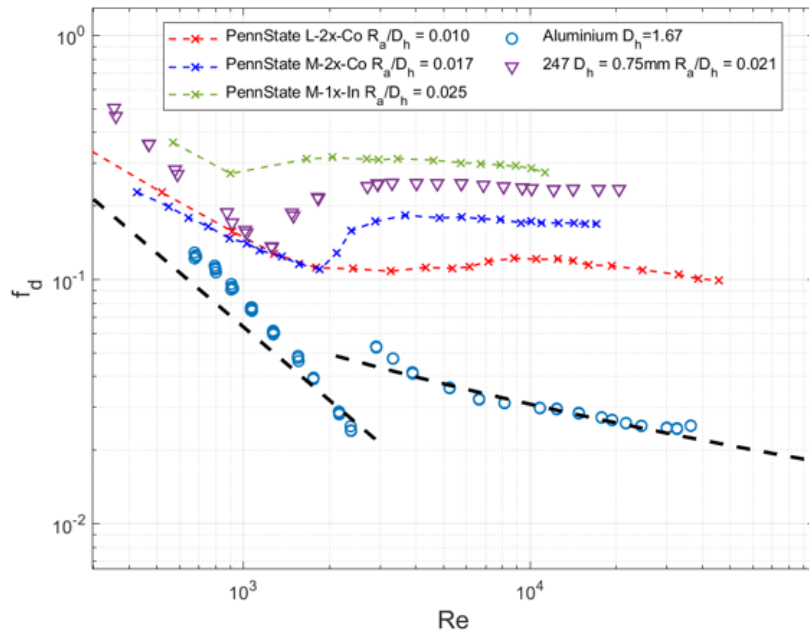


Figure 4.12: 247 channel with $D_h = 0.75\text{mm}$ comparison with PennState channels of similar relative roughness R_a/D_h

The 247 channel with $D_h = 0.75\text{mm}$ is presented together with results from PennState in Figure 4.12. The relative roughness of the 247 channels is 0.021 and the friction factor is located between the resulting friction factor of two of the channels from PennState that has a relative roughness of 0.017 and 0.025.

A deviation from the laminar line is present for the PennState results also indicating that this deviation can be due to the high relative roughness of the channels.

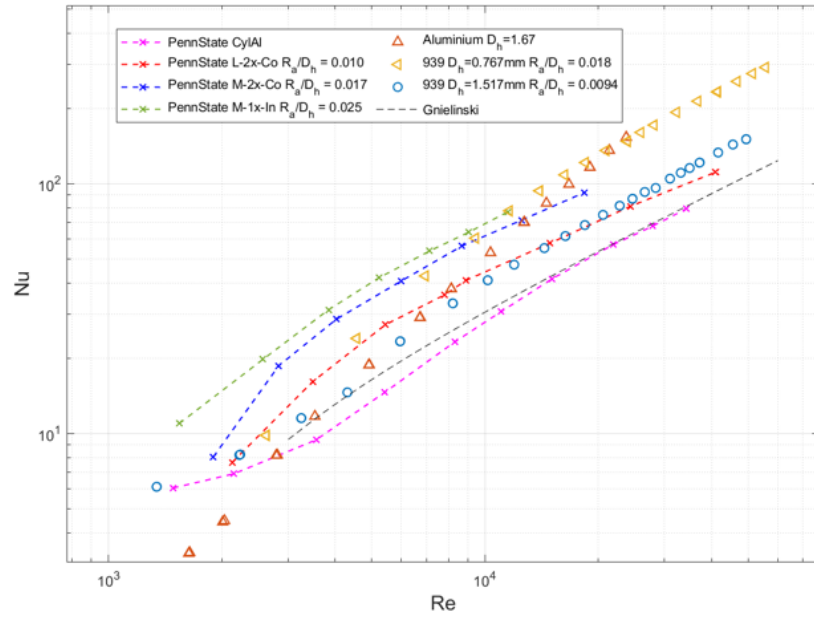
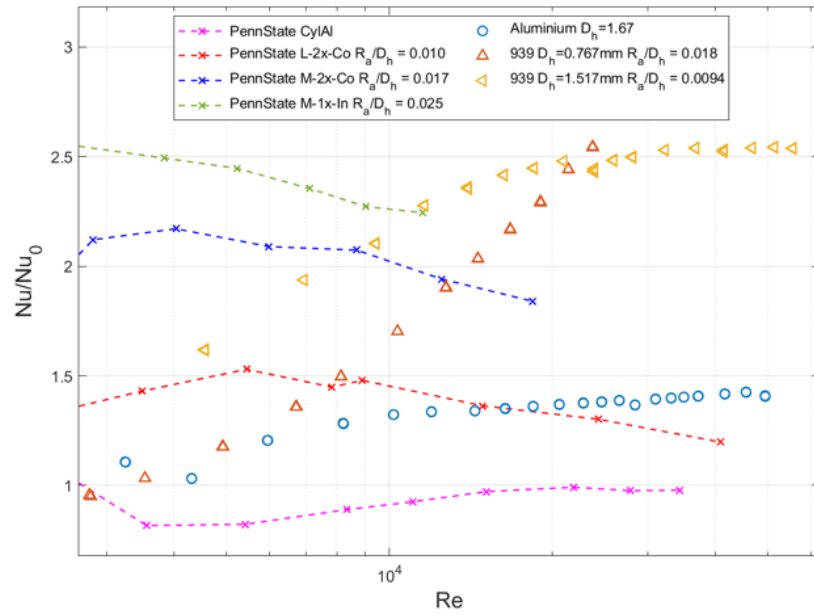


Figure 4.13: Nusselt number results comparison

Figure 4.14: Nu/Nu_0 results comparison

The Nusselt number results were compared between the two studies as well in Figure 4.13 and 4.14. Both studies show an increase of the heat transfer for the AM channels and that the heat transfer increases when the relative roughness R_a/D_h of the channels increased.

Discussion

5.1 Conclusions

This thesis has been a part of the work of examining new experimental methods for investigating the effects of an increased surface roughness in AM channels. Since AM has open up new possibilities within cooling design, it is necessary to perform experiments on the actual AM materials in order to gain more experience and knowledge about the materials and their properties.

The project has experimentally investigated the thermal performance and pressure losses in AM channels of two different materials, Inconel939 and CM247LC, by using a Steady State Heat Transfer rig which was assembled and validated. One channel had a rectangular shape while the rest were circular with different diameters (D) while the length was the same for all, namely $L=45$ mm. The Nusselt number and Darcy friction factor were used to evaluate the heat transfer and pressure losses of the channels.

The channels in this study had relatively low L/D ratios. Entrance effects were therefore most likely present and affecting the flow. This was evident during the validation process where the entrance effects were evaluated for the aluminium channels by applying a compensating factor to the Gnielinski correlation. Entrance effects should therefore be taken into consideration in future work with channels that have low L/D ratios.

The friction factor increased for all AM channels and converged towards a constant value for higher Reynolds numbers. This value increased with a decreasing hydraulic diameter and increased surface roughness, meaning that the friction factor increased when the relative roughness of the channels increased.

The Nusselt number did also increase with an increase of the relative roughness for all AM channels compared to smooth channels. The results also showed that the increase of the Nusselt number converged towards a constant value for higher Reynolds numbers.

Even though it was assumed that the roughness of the 939 material was greater than that of the 247 channel, this was not the case for the 247 channel with $D_h=0.75$ mm which had a roughness greater than the 939 channels. This was also shown in friction factor results where this channel yielded the highest friction factor. This indicates that it can not directly be assumed that AM channels of the same material have similar surface roughness.

The results were compared with results from PennState that have done similar studies on AM channels. The channels were compared by using the relative roughness in the form R_a/D_h . Channels with similar R_a/D_h resulted in similar friction factors for both studies, indicating that the R_a value

could be a potential parameter for estimating the friction factor of AM channels.

The obtained results can be used in the continued work of using AM components in cooling systems for industrial gas turbines. By knowing the thermal performance properties, these components can be used correctly in the design phase of the cooling systems.

5.2 Future work

Since the AM industry is fast paced, it is necessary to continue the work of examining the thermal performance of AM channels. The main focus of this thesis was about circular channels of different diameters of two different materials, while future work could include channels with a more variation in geometry. Rectangular channels with different aspect ratios could potentially be in focus for future work.

The length of the examined channels was 45mm, meaning that a test rig of this length was built and validated. A similar test rig of length 90mm is available at the Fluid Dynamic Lab at Siemens Energy. This rig needs to be assembled and validated and can then be used for similar investigation of thermal performance of test channels with length 90mm. Since the pressure drop increases for longer channels and the entrance effects becomes less significant, it is interesting to obtain results for longer channels.

Microscopic measurements were made to estimate the hydraulic diameter of the channels, however, these measurements were only made at the inlet and outlet of the channels. The dimensions of the channel further in of the channel are therefore not known and the estimation of the diameter could contain errors. A more precise method to estimate the hydraulic diameter would therefore improve the accuracy of the measurements. These measurements could potentially be done with a CT scan.

The surface roughness of the channels were measured in this project and the surface profile together with roughness values were obtained. For future work, it should be examined how these values can be used in the calculation instead of using the ε/D_h parameter.

The Nusselt number evaluation for low Reynolds numbers, or laminar flow, did not yield a constant value which was expected according to known correlation. It is suspected that the temperature measurement located at the outlet does not measure the correct outlet air temperature for very low mass flows. This should therefore be examined, either by simulations or experimentally, and a method for evaluating the Nusselt number at low Reynolds should be developed.

A new AM material has been produced, called Stal15. This material has a surface roughness somewhere between 247 and 939 and has proven potential being used in industrial gas turbines due to its high endurance regarding stresses in the material. The focus of a future project could therefore be about this new material and its behavior compared to the 247 and 939 materials.

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Appendix A

A.1 Matlab code

```
1 % Steady State Heat Transfer rig , main script
2
3 % Reading data from RigView. txt-file
4 NumberVariables = 30; % Number of variables in system
5 filename = '.txt';
6 coldtest = 0;
7 average = 1;
8 Al = 0;
9 rectangular = 0;
10
11 [A,sample_time ,P10 ,P11 ,P12 ,P13 ,P14 ,P15 ,P16 ,P4 ,P5 ,P6 ,P7 ,P8 ,P9 ,...
12 PT1 ,PT2 ,PT3 ,PT4 ,P_atm ,T1 ,T10 ,T11 ,T12 ,T2 ,T3 ,T4 ,T6 ,T9 ,...
13 W1,W2,mFlow] = readDataSSHT( filename , NumberVariables , average);
14
15 %%% Change D when changing test channel! %%%
16
17 D = 0.7862/1000; % diameter of channel [m]
18 r = D/2; % radius of channel [m]
19 A_c = pi*(r^2); % cross-section of channel [m^2]
20 P = pi*D; % perimeter P [m]
21
22 %%%
23
24 L = 0.045; % length of channel = 45 [mm]
25 A_s = P*L; % surface area [m]
26 Dh = 4*A_c/P; % hydraulic diameter [m]
27 mFlow = mFlow./1000; % massflow [kg/s]
28
29 if Al == 1
30 T_cu = (T1+T2+T3+T4)./4; % Thermocouple type-k measurements for copper
31 else
32 T_cu = (PT3+PT4)./2; % PT-100 measurements for copper
33 end
34 T_cu = T_cu + 273.15;
35 PT1 = PT1 + 273.15; % inlet
36 PT2 = PT2 + 273.15; % outlet
37
38 %%% Reynolds number %%%
39 mu0 = 1.716e-5; % Sutherlands law
40 T0=273.11;
41 S=110.56;
42 mu = mu0.*(PT1./T0).^1.5 * (T0+S)./(PT1+S);
43
44 Re_exp = (mFlow.*Dh)./(mu.*A_c);
45 [u-Re_exp] = UncertaintyReynolds(Dh, A_c, mFlow, PT1, mu);
```

```

46 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%% Nusselt (htc) %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
47 % Two different methods are used to calculate Nusselt number.
48 % Method 1: % Nusselt calculation when Twall = Tcu
49 C_p = 1005; Cv = 720; R = 8314/28.97;
50 k_air = mu.*Cv*(1.32+(1.77*(R./Cv)));
51
52 Q = mFlow.*C_p.*(PT2-PT1); % Q = m*C_p*(T_o-T_i) [W]
53 TLM = (PT2-PT1)./log((T_cu-PT1)./(T_cu-PT2));
54 h_1 = Q./(TLM.*A_s); % heat transfer coefficient [W/m^2/K]
55 Nu_exp = (h_1*Dh)./k_air; % Nusselt number for Twall = Tcu
56
57 % Method 2: Iterative process to calculate h, outlet temperature and wall
58 % temperature. Tests different h until right outlet temperature is achieved.
59 h = 800; h_hist = []; Nu = []; Nu_test = [];
60
61 if coldtest == 0
62     time = numel(sample_time);
63     for i = 1:time
64         T_copper = T_cu(i); T_inlet = PT1(i); T_outlet = PT2(i); mdot = mFlow(i);
65         Tdiff = 1;
66         q = Q(i);
67         while abs(Tdiff) > 0.10
68             [Tgas, Tmet, q1, Tvec, TmetIn, TmetOut, xvec] = getOutletTemperature(
69                 T_copper, T_inlet, mdot, h, Dh, rectangular);
70             Tdiff = T_outlet - Tgas;
71             if Tdiff > 0
72                 h = h*1.001;
73             else
74                 h = h*0.999;
75             end
76             h_hist(i) = h;
77             Nu(i) = (Dh*h)/k_air(i);
78             Tmet = Tmet + 273.15;
79             TLM_test(i) = (T_outlet-T_inlet)./log((Tmet-T_inlet)./(Tmet-T_outlet));
80             h_test = q./(TLM_test(i).*A_s);
81             Nu_test(i) = (h_test*Dh)./k_air(i);
82
83             [uNu(i), u_h(i), u_q(i), uTLM(i), uTw(i)] = UncertaintyNu(Dh, h, PT1(i),
84                 PT2(i), T_cu(i), mu(i), L, q, Tmet, TLM_test(i));
85         end
86     end
87 % Nusslet theorethical (Dittus-Boelter and Gnielinski)
88 Pr = 0.71; n = 0.4; % n = 0.4 for heated fluid. 0.3 for cooled fluid
89
90 Nu_DB = 0.023.*(Re_exp.^0.8).*(Pr.^n); % Dittus-Boelter equation for Nu.
91 f_Blasius = 0.3164.*(Re_exp.^-0.25); % Blasius friction factor.
92 Nu_G = ((f_Blasius./8).*(Re_exp-1000).*Pr)./(1+12.7*((f_Blasius./8).^(1/2)).*((Pr.^(2/3))-1)); % Gnielinski equation for Nusslet number. Turbulent flow in tubes.
93
94 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%% Friction factor %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
95 if coldtest == 1
96     P_inlet = P16;
97     P_inlet = P_inlet + P_atm;
98     P_outlet = P14;
99     P_outlet = P_outlet + P_atm;
100     P_diff = P9;
101
102 [DarcyF, DarcyF_diff, Mach, delta_p, plossin, plossout, p_tot1, p_tot2, p_tot_diff, rho_i,
    rho_o, u_DarcyF, u_DarcyF_diff, u_umean] = DarcyFrictionSHT(P_inlet, P_outlet, PT1,
    PT2, mFlow, Dh, A_c, L, P_diff, Re_exp, P16, P14, P_atm);
103 end

```

Listing A.1: Matlab code for main script

```

1 function [DarcyF, DarcyF_diff, Mach, delta_p, plossin, plossout, p_tot1, p_tot2,
2 p_tot_diff, rho_i, rho_o] = ...
3 DarcyFrictionSSHT(P_inlet, P_outlet, PT1, PT2, mFlow, Dh, A_c, L, P_diff, Re_exp, P16, P14
4 , P_atm)
5 % Calculates the Darcy friction factor
6 %
7 % The input:
8 % P_inlet: pressure at inlet [Pa]
9 % P_outlet: pressure at outlet [Pa]
10 % P_diff: pressure difference between inlet and outlet [Pa]
11 % mFlow: mass flow through the channel [kg/s]
12 % PT1: inlet temperature of air [K]
13 % PT2: outlet temperature of air [K]
14 % Dh: hydraulic diameter of channel [m]
15 % A_c: crosssection channel area [m2]
16 % L: channel length [m]
17 % Re_exp: Reynolds number [-]
18 %
19 % The output:
20 % DarcyF: Darcy friction factor
21 % DarcyF_diff: Darcy friction factor calculated with P_diff
22 % Mach: Mach number.
23 % delta_p:
24 % plossin/plossout: Pressure loss at inlet/outlet
25 % p_tot1/p_tot2: Total pressure at inlet/outlet
26 % p_tot_diff: Total pressure difference
27 % rho_i/rho_o: Density at inlet/outlet
28
29 M_air = 0.028969; % Molar mass [kg/mol]
30 R = 8.3145; % gas constant [Pa m^3/K/mol]
31
32 n = numel(Re_exp);
33 % Inlet(c1) and outlet(c2) loss coefficients for laminar and turbulent.
34 for j = 1:1:n
35     if Re_exp(j) > 1000 %Turbulent.
36         c1 = 0.5;
37         c2 = 1;
38     else
39         c1 = 0.5;
40         c2 = 2;
41     end
42
43 % Density at inlet and outlet
44 rho_i(j) = (P_inlet(j) * M_air) ./ (PT1(j) * R); %inlet
45 rho_o(j) = (P_outlet(j) * M_air) ./ (PT2(j) * R); %outlet
46
47 % Flow velocity
48 u_i(j) = mFlow(j) ./ (A_c * rho_i(j)); % inlet
49 u_o(j) = mFlow(j) ./ (A_c * rho_o(j)); % outlet
50
51 % Total pressure at inlet and outlet. p_tot = p_dyn + p_stat.
52 p_tot1(j) = P_inlet(j) + 0.5 * rho_i(j) * (u_i(j) ^ 2); % inlet
53 p_tot2(j) = P_outlet(j) + 0.5 * rho_o(j) * (u_o(j) ^ 2); % outlet
54 p_tot_diff(j) = p_tot1(j) - p_tot2(j);
55
56 % Mean density and velocity
57 rho_mean(j) = (rho_i(j) + rho_o(j)) ./ 2;
58 u_mean(j) = mFlow(j) ./ rho_mean(j) ./ A_c;
59
60 % Pressure loss at inlet and outlet
61 plossin(j) = c1 * 0.5 * mFlow(j) ^ 2 ./ rho_i(j) ./ (A_c ^ 2);
62 plossout(j) = c2 * 0.5 * mFlow(j) ^ 2 ./ rho_o(j) ./ (A_c ^ 2);

```

```

63 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%% Darcy Friction Factor %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
64
65 delta_p(j) = p_tot1(j)-c1*0.5*mFlow(j).^2./rho_i(j)./A_c.^2-p_tot2(j)-c2*0.5*mFlow(j)
66 .^2./rho_o(j)./A_c.^2;
67 delta_p_diff(j) = P_diff(j) + 0.5*rho_i(j).*(u_i(j).^2) - 0.5*rho_o(j).*(u_o(j).^2) -
68 c1*0.5*mFlow(j).^2./rho_i(j)./A_c.^2 - c2*0.5*mFlow(j).^2./rho_o(j)./A_c.^2;
69
70 DarcyF(j) = delta_p(j)./L*2./rho_mean(j)*Dh./u_mean(j).^2;
71 DarcyF_diff(j) = delta_p_diff(j)./L*2./rho_mean(j)*Dh./u_mean(j).^2;
72
73 % Mach number
74 u_o_Mach(j) = mFlow(j)./(A_c*rho_o(j));
75 Mach(j) = u_o_Mach(j)./343; % Highest velocity should be at the outlet
76
77 end
78 end
79

```

Listing A.2: Matlab function for calculation of Darcy friction factor

```

1 function [Tgas, Tmet, q, Tvec, TmetIn, TmetOut, xvec] = getOutletTemperature(T_copper, PT1
, mdot, h, Dh, rectangular)
2 %Function for calculating outlet temperature and wall temperature of test object. For now,
3 %assuming circular channel.
4 %
5 % The input:
6 % T_copper: Temperature of copper block [K]
7 % PT1: Inlet temperature of air
8 % massflow: Massflow through the channel [kg/s]
9 % h: the guessed heat transfer coefficient
10 % hcontact: heat transfer coefficient of thermal paste
11 % Dh: hydraulic diameter of test channel
12
13
14 L = 0.045; % length of channel
15 k_247 = 8.81; % thermal conductivity of SLM 247 [W/m/K]
16 k_939 = 10; %0.61; % thermal conductivity of SLM 939 [W/m/K]
17 Do = 10e-3; % test object outer diameter = 10 mm
18 Po = Do*pi;
19 Pi = Dh*pi;
20 ri = Dh/2; % inner radius
21 ro = Do/2; % outer radius
22 Cp = 1005;
23 hcontact = 1e4; %8.5/(0.08*10^-3); % thermal paste layer, #term pasta med gap om 0.02
mm MX-4: 8.5 W/m K
24
25
26 xvec = linspace(0, L, 100); % divide the channel into 100 cross sections
27 Tgas = PT1;
28 dx = xvec(2)-xvec(1);
29 Tvec = [];
30 TmetIn = [];
31 TmetOut = [];
32
33 i = 1;
34 for x = xvec %0:dx:L
35 if rectangular == 1
36 q = (Tgas - T_copper)/(1/(P_rect*h) + log(ro/ri)/(2*pi*k_247) + 1/(P_rect*hcontact
));
37 b = Tgas - (q/(P_rect*h));
38 Tgas = Tgas - ((q*dx)/(mdot*Cp));
39 a = -q/(2*pi*k_247);

```

```
40     else
41         q = (Tgas - T_copper)/(1/(Pi*h) + log(ro/ri)/(2*pi*k_247) + 1/(Po*hcontact)); %
         heat flux per meter pipe [W/m]
42         a = -q/(2*pi*k_247);
43         b = Tgas - q/(Pi*h); % inner metal temperature
44         Tgas = Tgas - ((q*dx)/(mdot*Cp)); % air temp
45     end
46
47     Tvec(i) = Tgas;
48     TmetIn(i) = b-273.15;
49     TmetOut(i) = (a*log(Do/Dh))+b-273.15;
50     i=i+1;
51 end
52
53 Tgas = Tvec(i-1);
54 Tmet = mean(TmetIn);
55
56 end
57
```

Listing A.3: Matlab function for calculation of outlet temperature for Nusselt calculations

A.2 Python code

```

1 import math, sys
2 import matplotlib.pyplot as plt
3 import numpy as np
4
5 D = 1.58e-3 # hydraulic diameter
6 A = 1.96e-6 #cross section flow area
7 Pi = 4.963e-3 # inner perimeter
8 hcontact = 1e4 #8.5/0.02e-3 #1e5 #term pasta med gap om 0.02 mm MX-4: 8.5 W/m K
9 L = 0.045 # meter length
10 OutputFile=".txt"
11
12 solveCases = [
13 {"Tin":21.633, "Tout":37.589, "Tcu":66.07, "mdot":1.17949, "Pin":7.24439572, "Pout":
14 :6.17852597},
15 {"Tin":21.634, "Tout":37.589, "Tcu":66.07, "mdot":1.17990, "Pin":7.24625787, "Pout":
16 :6.18117351},
17 {"Tin":21.56, "Tout":38.572, "Tcu":68.41, "mdot":1.14132, "Pin":7.25022728, "Pout":
18 :6.25744599},
19 ]
20
21 Cp = 1005. # J/kgK
22 plot = 0
23 def mu(T): #Sutherland's law
24     mu0 = 1.716e-5
25     T0=273.11
26     S=110.56
27     return mu0*math.pow(T/T0, 1.5) * (T0+S)/(T+S)
28 #
29 def rho(T,p): # T in Kelvin
30     R=8314/28.97
31     return p/(R*T)
32 #
33 def k(T): # T in Kelvin
34     Cv = 720
35     R=8314/28.97
36     return mu(T)*Cv*(1.32 + 1.77*R/Cv)
37 #
38 def getheatflux(Ti, To, ri, ro, hi, ho, Pi, Po, kmet):
39     return (Ti - To) / (1/(Pi*hi) + math.log(ro/ri)/(math.pi*2*kmet) + 1/(Po*ho)) # W/m
40     # heat flux per meter pipe
41 #
42 def entranchEnhance(xrD): # the accumulated increase of Nu up til x/D
43     return 1+8.7/(xrD+5)
44 #
45 def gnielinski(Re, Pr):
46     f = 0.3164/np.power(Re, 0.25)
47     return f/8*(Re-1000)*Pr/(1+12.7*np.sqrt(f/8)*(np.power(Pr, 2/3.)-1))
48 #
49 def getOutletTemperature(Tcopper, hcontact, Tinlet, mdot, h): #for now, assuming circular
50     # cross sect test objects
51     kmet = 8.81 #150 # W/mK Thermal conductivity for test object
52     Do = 10e-3 # SSHT rig test object outer diameter
53     Po = math.pi*Do
54
55     xvec = np.linspace(0, L, 100)
56     Tgas = Tinlet #
57     dx = xvec[1]-xvec[0]

```

```

56 Tvec = []
57 TmetIn = []
58 TmetOut = []
59
60 for x in xvec:
61     q = getheatflux(Tgas, Tcopper, D/2, Do/2, h, hcontact, Pi, Po, kmet)
62     a = -q / (2*math.pi*kmet)
63     b = Tgas - q/(Pi*h) # inner metal temperature
64 #     q = (Tgas - Tcopper) * Dinner*np.pi*h
65     Tgas -= q*dx/(mdot*Cp)
66     Tvec.append(Tgas-273.15)
67     TmetIn.append(b-273.15)
68     TmetOut.append(a*np.log(Do/D) + b -273.15)
69 #
70
71 if plot:
72     plt.plot(xvec,Tvec)
73     plt.plot(xvec,TmetIn)
74     plt.plot(xvec,TmetOut)
75     plt.plot(xvec,np.ones(len(xvec))*(Tcopper-273.15),'-')
76     plt.xlabel("Length [m]")
77     plt.ylabel("Gas temperature [C]")
78     plt.show()
79 #
80 return Tgas, np.average(TmetIn)+273.15 # outlet temperature
81 #
82
83 h = 1600
84 ReWrite=[]
85 NuWrite=[]
86 GnWrite=[]
87
88 for caseinfo in solveCases:
89     Tinlet = caseinfo["Tin"]+273.15
90     Toutlet = caseinfo["Tout"]+273.15
91     Tcopper = caseinfo["Tcu"]+273.15
92     mdot = caseinfo["mdot"]/1000
93     Pinlet = caseinfo["Pin"]*1e5
94     Poutlet = caseinfo["Pout"]*1e5
95
96     print("Mass flow: {0:.3f} g/s".format(mdot))
97     print("Inlet temperature: {0:.1f} C".format(Tinlet))
98     print("Outlet temperature: {0:.1f} C".format(Toutlet))
99
100     print("Inlet pressure: {0:.4f} bar".format(Pinlet))
101     print("Outlet pressure: {0:.4f} bar".format(Poutlet))
102
103     Tdiff = 1
104     print("Iterating to find htc...")
105     while np.abs(Tdiff) > .10:
106         Toutest, Tmet = getOutletTemperature(Tcopper, hcontact, Tinlet, mdot, h)
107         Tdiff = Toutlet - Toutest
108         if Tdiff > 0:
109             h *= 1.001
110         else:
111             h *= 0.999
112     plot = 0
113     getOutletTemperature(Tcopper, hcontact, Tinlet, mdot, h)
114     Re = mdot * D / (mu(Tinlet) * A )
115     print("Found HIC: {} W/m2K".format(h))
116     print("HIC (LMID): {}".format(mdot*Cp*(Toutlet-Tinlet)*np.log((Tmet-Tinlet)/(Tmet-Toutlet)) / (L*Pi*(Toutlet-Tinlet))))
117     print("Nusselt number: {}".format(D * h/k(Tinlet)))
118     print("Re: {}".format(Re))

```

```

119     print("Nusselt Gneilinski: {}".format(gnielinski(Re, 0.71)))
120
121     NuWrite.append(D * h/k(Tinlet))
122     GnWrite.append(gnielinski(Re, 0.71))
123
124     ### Friction factor
125
126     if Re < 2500:
127         inloss = 0.5
128         outloss = 2
129     else:
130         inloss = 0.5
131         outloss = 1
132
133     Ubulk_inlet = mdot/(rho(Tinlet, Pinlet) * A)
134     Ubulk_outlet = mdot/(rho(Toutlet, Poutlet) * A)
135
136     velheadIn = Ubulk_inlet**2*rho(Tinlet, Pinlet)/2
137     velheadOut = Ubulk_outlet**2*rho(Toutlet, Poutlet)/2
138     dp = Pinlet-Poutlet - outloss*velheadOut - inloss*velheadIn
139     print("Darcy friction factor: {}".format(dp*D/L / (0.5*(velheadIn+velheadOut))))
140     #
141     with open(OutputStream, "w") as outfile:
142         outfile.write("Reynolds\n")
143         for Re in ReWrite:
144             outfile.write("{}\n".format(Re))
145         outfile.write("Nusselt\n")
146         for dpn in NuWrite:
147             outfile.write("{}\n".format(dpn))
148         outfile.write("Gneilinski\n")
149         for dp1n in GnWrite:
150             outfile.write("{}\n".format(dp1n))
151
152

```

Listing A.4: Python code for heat flux in pipe calculations

```

1  import xml.etree.ElementTree as ET
2  import subprocess, sys
3
4  # Start C3D, then start task manager. In list of running programs, right-click in C3D ->
   Properties. There you'll find the path below
5  C3DExe =
6  '''
7  Create a C3D model of the test object
8  Add an extra "dummy" branch at the outlet (else the script cannot determine outflow
   temperature in the model.)
9  Make sure this branch do not cause pressure drop.
10 Make all branches the same:
11 Heat transfer: Custom model, with a reasonable value on C (here identical to nusselt
   number). n and s variables set to zero
12 Fanning friction factor - set a reasonable initial value.
13 LMID temperature mapping is most accurate - this choice affects reported nusselt number.
14 Mesh:
15 Couple flow network to mesh
16 Set outer diameter temperature and HTC. HTC will not be adjusted by the script below
17 The copper temperature will be applied to all external mesh faces that have a T and HTC
   defined.
18 This obviously includes mapped faces - but flow network will overwrite
19 When setting the case up, run the Conjugate solver to convergence, then save the model in
   that state.
20 This is the state C3D will run the case in batch mode later.
21 '''
22

```

```

23 solveCases = [
24 {"dir":"","name":"1.5mm","innode":7,"outnode":8,"upstreamnode":1,"Tin":22.16825, "
    Tout":60.824917, "Tcu":84.660333, "mdot":0.15079, "Pin":6.87875163, "Pout"
    :6.85601627},
25 {"dir":"","name":"1.5mm","innode":7,"outnode":8,"upstreamnode":1,"Tin":22.078417, "
    Tout":60.41875, "Tcu":84.139167, "mdot":0.20502, "Pin":6.63305472, "Pout"
    :6.58880632},
26 ]
27
28 OutputFile=".txt"
29 tol_mdot = 0.001 # g/s
30 tol_temp = 0.02 # Celcius
31 DarcyWrite=[]
32 NuWrite=[]
33
34 for caseinfo in solveCases:
35     dir = caseinfo["dir"]
36     name = caseinfo["name"]
37     innode = caseinfo["innode"]
38     outnode = caseinfo["outnode"]
39     Tinlet = caseinfo["Tin"]
40     Toutlet = caseinfo["Tout"]
41     Tcopper = caseinfo["Tcu"]
42     mdot = caseinfo["mdot"]
43     Pinlet = caseinfo["Pin"]
44     Poutlet = caseinfo["Pout"]
45     upstreamoutletnode = caseinfo["upstreamnode"]
46
47     print("Processing case {}/{}.sas".format(dir,name))
48     print("Mass flow: {0:.3f} g/s".format(mdot))
49     print("Inlet temperature: {0:.1f} C".format(Tinlet))
50     print("Outlet temperature: {0:.1f} C".format(Toutlet))
51     print("Copper temperature: {0:.1f} C".format(Tcopper))
52     print("Inlet pressure: {0:.4f} bar".format(Pinlet))
53     print("Outlet pressure: {0:.4f} bar".format(Poutlet))
54     modelFile = dir+"/"+name+".sas"
55
56     # Set up flow network according to measurements
57     FlowNetXML = ET.parse(dir+"/"+name+"_ModelData/FlowNetwork.xml")
58     FlowNetXMLroot = FlowNetXML.getroot()
59
60     for entry1 in FlowNetXMLroot:
61         if entry1.tag == "XMLNodeValues":
62             for guinode in entry1:
63                 for stuff in guinode:
64                     if stuff.tag == "FlowNode":
65                         setprops = False
66                         for nodestuff in stuff:
67                             if nodestuff.tag == "NodeNr":
68                                 if int(nodestuff.text) == innode:
69                                     setprops=True
70                                     setpressure = Pinlet*1e5
71                                     settemperature = Tinlet+273.15
72                                 elif int(nodestuff.text) == outnode:
73                                     setprops=True
74                                     setpressure = Poutlet*1e5
75                                     settemperature = Toutlet+273.15
76                             if nodestuff.tag == "Pressure" and setprops:
77                                 nodestuff.text = str(setpressure)
78                             if nodestuff.tag == "Temp" and setprops:
79                                 nodestuff.text = str(settemperature)
80
81     #
82     # Check values on Nu and friction factor...
83     if entry1.tag == "XMLBranchValues":

```

```

83         for guibranch in entry1:
84             for stuff in guibranch:
85                 if stuff.tag == "FlowBranch":
86                     for branchstuff in stuff:
87                         if branchstuff.tag == "friction":
88                             Fanning = float(branchstuff.text)
89                         if branchstuff.tag == "Correlation":
90                             for corrrstuff in branchstuff:
91                                 if corrrstuff.tag == "C":
92                                     Nusselt = float(corrrstuff.text)
93
94     # Write the updated XML file
95     FlowNetXML.write(dir+"/"+name+"_ModelData/FlowNetwork.xml")
96
97     # Set copper temperature. Do this on all faces that have a recovery temperature set
98     # Flow network will overwrite this for coupled faces
99     # Adiabatic faces will not be set (They do not have a "Tr" tag)
100    tinfXML = ET.parse(dir+"/"+name+"_Transient/t_inf.xml")
101    tinfXMLroot = tinfXML.getroot()
102
103    for entry1 in tinfXMLroot:
104        if entry1.tag == "MeshData":
105            for facedata in entry1:
106                for prop in facedata:
107                    if prop.tag == "Tr":
108                        prop.text = str(Tcopper+273.15)
109        if entry1.tag == "NetworkNodeData":
110            for nodeinfo in entry1:
111                settandp = False
112                for stuff in nodeinfo:
113                    if stuff.tag == "N":
114                        if int(stuff.text) == innode:
115                            settandp = True
116                            ttoiset = Tinlet
117                            ptoset = Pinlet
118                        if int(stuff.text) == outnode:
119                            settandp = True
120                            ttoiset = Toutlet
121                            ptoset = Poutlet
122                    if stuff.tag == "T" and settandp:
123                        stuff.text = str(ttoiset)
124                    if stuff.tag == "P" and settandp:
125                        stuff.text = str(ptoset)
126    tinfXML.write(dir+"/"+name+"_Transient/t_inf.xml")
127
128    notmatched = True
129    while notmatched:
130        print("Running C3D...")
131
132        subprocess.call([C3DExe, modelFile, '-solve', '-save'], stdin=None, stdout=None,
133                        stderr=None, shell=False)
134        tinfXML = ET.parse(dir+"/"+name+"_Transient/t_inf.xml")
135        tinfXMLroot = tinfXML.getroot()
136
137        for entry1 in tinfXMLroot:
138            if entry1.tag == "NetworkBranchData":
139                for branchdata in entry1[0]:
140                    if branchdata.tag == "m":
141                        modelmdot = float(branchdata.text)
142            if entry1.tag == "NetworkNodeData":
143                for nodeinfo in entry1:
144                    gettemp = False
145                    for stuff in nodeinfo:
146                        if stuff.tag == "N":

```

```

146         if int(stuff.text) == upstreamoutletnode:
147             gettemp = True
148             if stuff.tag == "T" and gettemp:
149                 modeloutlettemp = float(stuff.text)
150             print("Model massflow: {0:.5f} measured: {1:.3f} g/s. Model outlet temperature
151                   {2:.1f} measured: {3:.1f} C".format(modelmdot, mdot, modeloutlettemp, Toutlet))
152             if abs(modelmdot - mdot) < tol_mdot and abs(modeloutlettemp - Toutlet) < tol_temp:
153                 notmatched = False
154                 print("Case converged. Fanning friction factor: {0:.4f} Darcy: {1:.4f},
155                       Nusselt number: {2:.2f}".format(Fanning, Fanning*4, NusseltLast))
156                 DarcyWrite.append(Fanning*4)
157                 NuWrite.append(NusseltLast)
158                 break
159             if modelmdot > mdot:
160                 fricadjust = 1+1*abs(modelmdot - mdot)/mdot
161             else:
162                 fricadjust = max(0.3, 1-1*abs(modelmdot - mdot)/mdot)
163             if modeloutlettemp > Toutlet:
164                 htcadjust = max(0.1, 1-1*abs(modeloutlettemp - Toutlet)/Toutlet)
165             else:
166                 htcadjust = min(1+1*abs(modeloutlettemp - Toutlet)/Toutlet, 4)
167
168             FlowNetXML = ET.parse(dir+"/"+name+"_ModelData/FlowNetwork.xml")
169             FlowNetXMLroot = FlowNetXML.getroot()
170
171             for entry1 in FlowNetXMLroot:
172                 if entry1.tag == "XMLBranchValues":
173                     for guibranch in entry1:
174                         for stuff in guibranch:
175                             if stuff.tag == "FlowBranch":
176                                 for branchstuff in stuff:
177                                     if branchstuff.tag == "friction":
178                                         branchstuff.text = str(float(branchstuff.text)*
179                                                                   fricadjust)
180                                     if branchstuff.tag == "Correlation":
181                                         for corrstuff in branchstuff:
182                                             if corrstuff.tag == "C":
183                                                 NusseltLast = Nusselt
184                                                 corrstuff.text = str(float(corrstuff.text)*
185                                                                   htcadjust)
186                                     Nusselt = float(corrstuff.text)
187
188             #
189             FlowNetXML.write(dir+"/"+name+"_ModelData/FlowNetwork.xml")
190
191 #
192 with open(OutputStream, "w") as outfile:
193     outfile.write("Darcy\n")
194     for Da in DarcyWrite:
195         outfile.write("{}\n".format(Da))
196
197     for dpn in NuWrite:
198         outfile.write("{}\n".format(dpn))

```

Listing A.5: Python code for heat transfer calculations using a 3D model in the in-house simulation program C3D

Appendix B

B.1 Data sheets



REPORT

issued by an Accredited Testing Laboratory

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Date
2017-12-07
Reference
7P08259

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1 (4)



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Thermal conductivity of sintered material at different temperatures

(1 appendix)

This report details the specimens, the test conditions and the results obtained as the thermal conductivity, λ , in different directions (axial and radial) was determined following the procedure outlined in the accredited test method ISO 22007-2:2008, Transient Plane hot Source (TPS). The tests depart from the standard in the specimens not being plastics.

The main results are given below.

Temp. (°C)	λ_{ax} (W/m·K)	λ_{rad} (W/m·K)
22	7.5	8.2
55	8.5	8.8
80	8.9	9.2

Product designation

One types of sintered material was sent to RISE in November 2017. No designation was found on the material.

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REPORT

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1 Purpose of the test

The purpose of the test was to determine the thermal conductivity (λ) both in-plane and axial to the cylindrical symmetry axes in the material at three different temperatures.

2 Test specimens

Three cylinders were sent to RISE. They were 50 mm in diameter and 50 mm high. From one of the cylinders a small disc, 20 mm in diameter and 5.5 mm high was cut. This cylinder was used for heat capacity testing to be used as an input to the thermal conductivity measurements.

3 Measurements

3.1 Density, ρ

The density was measured on all three cylinders (before cutting one of them) using a 0.1 μg resolution balance and a calliper.

3.2 Heat capacity, c_p

Specific heat capacity of the samples was determined using the sample holder designated for specific heat capacity measurements Using TPS equipment. It is a brass cylinder with 20 mm diameter and 6 mm height attached to a TPS sensor of type 5501 and radius 6.4 mm. The holder is well insulated in polystyrene insulation. The procedure follows that described in the Hot Disc Thermal Constants Analyzer Manual (revision date 2014-03-27). Five measurements per temperature were performed. The accuracy of the method should be within 7 % according to the manufacturer of the equipment. Table 1 summarises the experimental parameters.

Table 1. Experimental parameters of the heat capacity measurements.

Temperature ($^{\circ}\text{C}$)	Measurement	Power (mW)	Meas. Time (s)	Mass (g)
22	Reference	90	80	0
	Sample	350		15.91
55	Reference	90		0
	Sample	350		15.91
80	Reference	90		0
	Sample	350		15.91

3.3 Anisotropic thermal conductivity, λ

The thermal conductivity measurements (in both radial (in-plane) and axial (through-plane) direction) were performed according to the ISO 22007-2:2008 standard using a TPS 2500 S and the so called anisotropy module. The sensor of type 5501 had a 25 μm Teflon insulating coating and a radius of 6.4 mm. At least 12 tests were performed at each temperature.

The analysis is done with predefined volumetric heat capacity (ρc_p) as defined from the results of the tests described above. The density is assumed temperature independent within the interval studied. In addition, variation of the results were analysed using different analysis interval of the transient (starting from point 30 to point 80 and ending from point 160 to

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point 200), the margin of error for the heat capacity measurements and the standard deviation of the tests performed. Table 2 summarises the experimental parameters.

Table s. Experimental parameters of the anisotropic thermal conductivity measurements.

Parameter	Value
Sensor insulation	Kapton
Insulation thickness	25 μm
Sensor type	5501
Sensor radius	6.4 mm
Heating power	750 mW
Measuring time	10 s
Point analysed (for main result)	50-200
Time between consecutive measurements	50 min
Applied external force	5 N
Temperatures	22, 50 and 80 $^{\circ}\text{C}$
Data points collected during transient	200

No unexpected events were observed.

4 Results

4.1 Density

The density was determined to 8600 kg/m^3 with a total spread between the three cylinders of 0.1 %.

4.2 Specific heat capacity

The results of the specific heat capacity are shown in Table 3. All measurements are shown in the appendix.

Table 3. Specific heat capacity measurements of the material.

Temperature ($^{\circ}\text{C}$)	c_p (J/kgK)	Point in analysis
22	413	66-132
55	433	65-130
80	442	56-112



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4.3 Anisotropic thermal conductivity

The results of the analysis for anisotropic thermal conductivity are shown in Table 4.

Table 4. Results of the anisotropic thermal conductivity analysis. Values in the parenthesis are margins of error, taken into account the margin of error in the input data (heat capacity).

Temp. (°C)	ρc_p (MJ/m ³ K)	λ_{ax} (W/m·K)	λ_{rad} (W/m·K)	Radial probing depth (mm)	Anisotropy (radial/axial)
22	3.55	7.51 (0.76)	8.19 (0.83)	9.6	109%
55	3.72	8.51 (1.13)	8.81 (1.20)	9.7	104%
80	3.80	8.85 (1.03)	9.24 (1.05)	9.9	104%

The thermal conductivity of the material is almost isotropic. It should be noted that the major part of the margin of error comes from the total margin of error in the heat capacity used as input to the analysis. Changing the input by 7 % results in just less than 7 % change in the conductivities. Varying the analysis interval results in less than 1 % variation and the standard deviation between measurements are mostly less than 2 %.

RISE Research Institutes of Sweden AB
Safety - Fire Research, Fire Resistance

Performed by

Johan Sjöström

Examined by

Signed by: Patrik Johansson
Reason: I have reviewed this document
Date & Time: 2017-12-08 08:34:39 +01:00
Patrik Johansson

Appendix

RISE Research Institutes of Sweden AB



REPORT

Date
2017-12-07Reference
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Appendix 1

Table A1. Individual measurements of heat capacity at 22 °C

Sample ID	(Points)	Temperature	Sp.Heat Cap.	Sp. Heat	Temp.Incr. (ref.)	Temp.Incr. (w. sample)
Prov Siemens 9mW 80s RT	(66-132)	22.0 °C	6.4681 J/K	406.60 J/KgK	0.218 K	0.124 K
Prov Siemens 9mW 80s RT	(66-132)	22.0 °C	6.5797 J/K	413.61 J/KgK	0.218 K	0.123 K
Prov Siemens 9mW 80s RT	(66-132)	22.0 °C	6.6214 J/K	416.23 J/KgK	0.218 K	0.122 K
Prov Siemens 9mW 80s RT	(66-132)	22.0 °C	6.6162 J/K	415.90 J/KgK	0.218 K	0.122 K
Average (3:6)		22.0 °C	6.5714 J/K	413.09 J/KgK	0.218 K	0.123 K
StandardDeviation (3:6)		0.0 °C	0.0617 J/K	3.88 J/KgK	0.000 K	0.001 K

Table A2. Individual measurements of heat capacity at 55 °C

Sample ID	(Points)	Temperature	Sp.Heat Cap.	Sp. Heat	Temp.Incr. (ref.)	Temp.Incr. (w. sample)
Prov Siemens 350mW 80s 55°C	(65-130)	55.0 °C	6.8426 J/K	430.14 J/KgK	2.01 K	1.15 K
Prov Siemens 350mW 80s 55°C	(65-130)	55.0 °C	6.8516 J/K	430.70 J/KgK	2.01 K	1.15 K
Prov Siemens 350mW 80s 55°C	(65-130)	55.0 °C	6.8971 J/K	433.56 J/KgK	2.01 K	1.15 K
Prov Siemens 350mW 80s 55°C	(65-130)	55.0 °C	6.9283 J/K	435.53 J/KgK	2.01 K	1.14 K
Prov Siemens 350mW 80s 55°C	(65-130)	55.0 °C	6.9030 J/K	433.93 J/KgK	2.01 K	1.15 K
Average (1:5)		55.0 °C	6.8845 J/K	432.77 J/KgK	2.01 K	1.15 K
StandardDeviation (1:5)		0.0 °C	0.0324 J/K	2.04 J/KgK	0.00 K	0.00 K

Table A3. Individual measurements of heat capacity at 80 °C

Sample ID	(Points)	Temperature	Sp.Heat Cap.	Sp. Heat	Temp.Incr. (ref.)	Temp.Incr. (w. sample)
Prov Siemens 350mW 80s 80°C	(56-112)	80.0 °C	7.0488 J/K	443.10 J/KgK	1.73 K	0.972 K
Prov Siemens 350mW 80s 80°C	(56-112)	80.0 °C	7.0193 J/K	441.25 J/KgK	1.73 K	0.975 K
Prov Siemens 350mW 80s 80°C	(56-112)	80.0 °C	7.0096 J/K	440.63 J/KgK	1.73 K	0.977 K
Prov Siemens 350mW 80s 80°C	(56-112)	80.0 °C	7.0566 J/K	443.59 J/KgK	1.73 K	0.971 K
Prov Siemens 350mW 80s 80°C	(56-112)	80.0 °C	6.9866 J/K	439.19 J/KgK	1.73 K	0.980 K
Average (3:7)		80.0 °C	7.0242 J/K	441.55 J/KgK	1.73 K	0.975 K
StandardDeviation (3:7)		0.0 °C	0.0257 J/K	1.62 J/KgK	0.00 K	0.003 K

Table A4. Individual measurements of thermal conductivity 22 °C

Sample ID	(Points)	Temperature	Ax.Th.Conductivity	Ax.Th.Diffusivity	Rad.Th.Conductivity	Rad.Th.Diffusivity	Ax.Pr.Depth	Rad.Pr.Depth
Siemens 5501 rdd (1) 750mW 10s	(50-200,ic)	22.0 °C	7.4145 W/mK	2.0866 mm²/s	8.3563 W/mK	2.3516 mm²/s	9.14 mm	9.70 mm
Siemens 5501 rdd (1) 750mW 10s	(50-200,ic)	22.0 °C	7.3906 W/mK	2.0799 mm²/s	8.2587 W/mK	2.3242 mm²/s	9.12 mm	9.64 mm
Siemens 5501 rdd (1) 750mW 10s	(50-200,ic)	22.0 °C	7.6882 W/mK	2.1636 mm²/s	8.0703 W/mK	2.2711 mm²/s	9.30 mm	9.53 mm
Siemens 5501 rdd (1) 750mW 10s	(50-200,ic)	22.0 °C	7.6668 W/mK	2.1576 mm²/s	8.0771 W/mK	2.2731 mm²/s	9.29 mm	9.54 mm
Siemens 5501 rdd (1) 750mW 10s	(50-200,ic)	22.0 °C	7.5539 W/mK	2.1258 mm²/s	8.1582 W/mK	2.2969 mm²/s	9.22 mm	9.58 mm
Siemens 5501 rdd (1) 750mW 10s	(50-200,ic)	22.0 °C	7.6194 W/mK	2.1443 mm²/s	8.1115 W/mK	2.2828 mm²/s	9.26 mm	9.56 mm
Siemens 5501 rdd (1) 750mW 10s	(50-200,ic)	22.0 °C	7.5985 W/mK	2.1384 mm²/s	8.1226 W/mK	2.2859 mm²/s	9.25 mm	9.56 mm
Siemens 5501 rdd (1) 750mW 10s	(50-200,ic)	22.0 °C	7.4072 W/mK	2.0845 mm²/s	8.2499 W/mK	2.3217 mm²/s	9.13 mm	9.64 mm
Siemens 5501 rdd (1) 750mW 10s	(50-200,ic)	22.0 °C	7.5067 W/mK	2.1125 mm²/s	8.1865 W/mK	2.3038 mm²/s	9.19 mm	9.60 mm
Siemens 5501 rdd (1) 750mW 10s	(50-200,ic)	22.0 °C	7.4556 W/mK	2.0981 mm²/s	8.2143 W/mK	2.3117 mm²/s	9.16 mm	9.62 mm
Siemens 5501 rdd (1) 750mW 10s	(50-200,ic)	22.0 °C	7.6480 W/mK	2.1523 mm²/s	8.0877 W/mK	2.2761 mm²/s	9.28 mm	9.54 mm
Siemens 5501 rdd (1) 750mW 10s	(50-200,ic)	22.0 °C	7.2970 W/mK	2.0535 mm²/s	8.3159 W/mK	2.3403 mm²/s	9.06 mm	9.68 mm
Siemens 5501 rdd (1) 750mW 10s	(50-200,ic)	22.0 °C	7.3985 W/mK	2.0824 mm²/s	8.2471 W/mK	2.3209 mm²/s	9.13 mm	9.64 mm
Average (3:15)		22.0 °C	7.5112 W/mK	2.1138 mm²/s	8.1890 W/mK	2.3045 mm²/s	9.19 mm	9.60 mm
StandardDeviation (3:15)		0.0 °C	0.1214 W/mK	0.0342 mm²/s	0.0898 W/mK	0.0293 mm²/s	0.07 mm	0.05 mm

Table A5. Individual measurements of thermal conductivity 55 °C

Sample ID	(Points)	Temperature	Ax.Th.Conductivity	Ax.Th.Diffusivity	Rad.Th.Conductivity	Rad.Th.Diffusivity	Ax.Pr.Depth	Rad.Pr.Depth
Siemens 5501 rdd (1) 750mW 10s 55°C	(50-200,ic)	55.0 °C	8.8877 W/mK	2.3874 mm²/s	8.4540 W/mK	2.2858 mm²/s	9.77 mm	9.82 mm
Siemens 5501 rdd (1) 750mW 10s 55°C	(50-200,ic)	55.0 °C	8.3510 W/mK	2.2433 mm²/s	8.9391 W/mK	2.4012 mm²/s	9.47 mm	9.80 mm
Siemens 5501 rdd (1) 750mW 10s 55°C	(50-200,ic)	55.0 °C	8.4650 W/mK	2.2747 mm²/s	8.7854 W/mK	2.3599 mm²/s	9.54 mm	9.72 mm
Siemens 5501 rdd (1) 750mW 10s 55°C	(50-200,ic)	55.0 °C	8.4960 W/mK	2.2822 mm²/s	8.7281 W/mK	2.3446 mm²/s	9.55 mm	9.68 mm
Siemens 5501 rdd (1) 750mW 10s 55°C	(50-200,ic)	55.0 °C	8.4969 W/mK	2.2825 mm²/s	8.8645 W/mK	2.3812 mm²/s	9.56 mm	9.76 mm
Siemens 5501 rdd (1) 750mW 10s 55°C	(50-200,ic)	55.0 °C	8.2614 W/mK	2.2246 mm²/s	8.9856 W/mK	2.4141 mm²/s	9.43 mm	9.53 mm
Siemens 5501 rdd (1) 750mW 10s 55°C	(50-200,ic)	55.0 °C	8.8270 W/mK	2.3980 mm²/s	8.5063 W/mK	2.2850 mm²/s	9.73 mm	9.56 mm
Siemens 5501 rdd (1) 750mW 10s 55°C	(50-200,ic)	55.0 °C	8.3917 W/mK	2.2542 mm²/s	8.9211 W/mK	2.3964 mm²/s	9.50 mm	9.79 mm
Siemens 5501 rdd (1) 750mW 10s 55°C	(50-200,ic)	55.0 °C	8.2941 W/mK	2.2280 mm²/s	8.9813 W/mK	2.4126 mm²/s	9.44 mm	9.82 mm
Siemens 5501 rdd (1) 750mW 10s 55°C	(50-200,ic)	55.0 °C	8.9108 W/mK	2.3937 mm²/s	8.4919 W/mK	2.2811 mm²/s	9.79 mm	9.55 mm
Siemens 5501 rdd (1) 750mW 10s 55°C	(50-200,ic)	55.0 °C	8.3892 W/mK	2.2535 mm²/s	8.8678 W/mK	2.3575 mm²/s	9.49 mm	9.77 mm
Siemens 5501 rdd (1) 750mW 10s 55°C	(50-200,ic)	55.0 °C	8.1683 W/mK	2.1942 mm²/s	9.1593 W/mK	2.4604 mm²/s	9.37 mm	9.92 mm
Average (3:14)		55.0 °C	8.5052 W/mK	2.2847 mm²/s	8.8071 W/mK	2.3658 mm²/s	9.56 mm	9.73 mm
StandardDeviation (3:14)		0.0 °C	0.2497 W/mK	0.0671 mm²/s	0.2165 W/mK	0.0582 mm²/s	0.14 mm	0.12 mm



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Appendix 1

Table A6. Individual measurements of thermal conductivity 80 °C

Sample ID	(Points)	Temperature	Air Th. Conductivity	Air Th. Diffusivity	Radi Th. Conductivity	Radi Th. Diffusivity	Air Pr. Depth	Radi Pr. Depth
Siemens 5501 rdd (1) 750mW 10s 80°C	(50-200,10)	80.0 °C	8.9773 W/mK	2.3636 mm²/s	9.0940 W/mK	2.3943 mm²/s	9.72 mm	9.79 mm
Siemens 5501 rdd (1) 750mW 10s 80°C	(50-200,10)	80.0 °C	9.2345 W/mK	2.4313 mm²/s	8.9376 W/mK	2.3531 mm²/s	9.86 mm	9.70 mm
Siemens 5501 rdd (1) 750mW 10s 80°C	(50-200,10)	80.0 °C	8.6832 W/mK	2.2861 mm²/s	8.3363 W/mK	2.4581 mm²/s	9.56 mm	9.92 mm
Siemens 5501 rdd (1) 750mW 10s 80°C	(50-200,10)	80.0 °C	8.9252 W/mK	2.3498 mm²/s	9.2472 W/mK	2.4346 mm²/s	9.70 mm	9.87 mm
Siemens 5501 rdd (1) 750mW 10s 80°C	(50-200,10)	80.0 °C	8.3491 W/mK	2.1982 mm²/s	9.6386 W/mK	2.5377 mm²/s	9.38 mm	10.1 mm
Siemens 5501 rdd (1) 750mW 10s 80°C	(50-200,10)	80.0 °C	8.8953 W/mK	2.3420 mm²/s	9.1900 W/mK	2.4196 mm²/s	9.68 mm	9.84 mm
Siemens 5501 rdd (1) 750mW 10s 80°C	(50-200,10)	80.0 °C	9.1484 W/mK	2.4086 mm²/s	9.9853 W/mK	2.3658 mm²/s	9.82 mm	9.73 mm
Siemens 5501 rdd (1) 750mW 10s 80°C	(50-200,10)	80.0 °C	8.8024 W/mK	2.3175 mm²/s	9.1910 W/mK	2.4198 mm²/s	9.63 mm	9.84 mm
Siemens 5501 rdd (1) 750mW 10s 80°C	(50-200,10)	80.0 °C	9.1810 W/mK	2.4172 mm²/s	9.0091 W/mK	2.3719 mm²/s	9.83 mm	9.74 mm
Siemens 5501 rdd (1) 750mW 10s 80°C	(50-200,10)	80.0 °C	8.5406 W/mK	2.2486 mm²/s	9.5084 W/mK	2.5034 mm²/s	9.48 mm	10.0 mm
Siemens 5501 rdd (1) 750mW 10s 80°C	(50-200,10)	80.0 °C	8.9428 W/mK	2.3545 mm²/s	9.1673 W/mK	2.4126 mm²/s	9.70 mm	9.83 mm
Siemens 5501 rdd (1) 750mW 10s 80°C	(50-200,10)	80.0 °C	8.4803 W/mK	2.2327 mm²/s	9.5340 W/mK	2.5101 mm²/s	9.45 mm	10.0 mm
Average (3/14)		80.0 °C	8.8467 W/mK	2.3292 mm²/s	9.2366 W/mK	2.4318 mm²/s	9.65 mm	9.86 mm
StandardDeviation (3/14)		0.0 °C	0.2727 W/mK	0.0718 mm²/s	0.2174 W/mK	0.0572 mm²/s	0.15 mm	0.12 mm

SIEMENS

CALIBRATION PROTOCOL

Kalibreringsprotokoll

Number (nummer)	Page (of)/Sida av
122087_91	1(2)
Date of Calibration (Kalibreringsdatum)	
2019-03-08	

PRESSURE SCANNER MODULE

Object Manufacturer: Pressure Systems Inc. Type: 9116 TC122087
Condition of the object: Good.
Object IP.Adr: 200.200.23.232
Object firmware updated to v7.02 before calibration.

Customer Siemens Industrial Turbomachinery AB, SE-612 83 Finspong, Sweden.

Standards Pressure Calibrator, Fluke 6270A TC128393
Pressure Calibrator, Druck PACE 5000 (± 2.5 psi) TC121845

Traceability The laboratory reference standards for pressures and electrical quantities are through regularly performed calibrations directly traceable to the Research Institutes of Sweden, RISE in Borås.

Method of Calibration The object was calibrated with the standards at atmosph. pressure.
The measured value was logged in the computer using the "PSI 9816 Calibration software".
The measured value is a arithmetic mean of at least three readings

Calibration uncertainty The reported expanded uncertainty of measurement is stated as the standard uncertainty of measurement multiplied by the coverage factor $k=2$ which for a normal distribution corresponds to a coverage probability of approximately 95%. The standard uncertainty of measurement has been determined in accordance with EAL Publication EA-4/02. The expanded uncertainty includes estimated uncertainties from all quantities that have been evaluated to influence the measurement.

Ch.number	Expanded uncertainty [% of applied pressure]
1 - 6	0.1
7 - 16	0.01

Calibration Conditions Ambient Temperature: 23 ± 2 °C Warm-up Time: > 24 h
Atmospheric Pressure: 975.8 hPa Pressure medium: Nitrogen

The result of this calibration is valid for the calibrated object at the date of calibration.
Measurement of long time instability is not performed.

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TC122087
2019-03-08Page (of)/Sida av
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Results

Applied (psi)	Reading ch. #1 (psi)	# 2 (psi)	# 3 (psi)	# 4 (psi)	# 5 (psi)	# 6 (psi)
-2.500	-2.501	-2.500	-2.500	-2.501	-2.501	-2.501
-2.000	-2.003	-2.003	-2.003	-2.003	-2.003	-2.003
-1.500	-1.503	-1.503	-1.503	-1.503	-1.503	-1.503
-1.000	-1.003	-1.003	-1.003	-1.003	-1.003	-1.003
-0.500	-0.501	-0.501	-0.501	-0.501	-0.501	-0.501
0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.500	0.502	0.502	0.501	0.502	0.502	0.502
1.000	1.003	1.002	1.003	1.003	1.003	1.003
1.500	1.504	1.503	1.503	1.504	1.504	1.504
2.000	2.003	2.002	2.002	2.003	2.003	2.003
2.500	2.502	2.500	2.500	2.501	2.501	2.501

Applied (psi)	Reading ch. # 7 (psi)	# 8 (psi)
-10.000	-10.000	-10.000
-8.000	-8.001	-8.001
-6.000	-6.000	-6.000
-4.000	-4.000	-4.000
-2.000	-2.000	-2.000
0.000	-0.001	-0.001
2.000	2.000	2.000
4.000	4.000	4.000
6.000	5.999	6.000
8.000	8.000	8.000
10.000	9.998	9.999

Applied (psi)	Reading ch. # 9 (psi)	# 10 (psi)
-15.000	-15.012	-15.002
-12.000	-12.009	-12.002
-9.000	-9.007	-9.002
-6.000	-6.005	-6.001
-3.000	-3.002	-3.002
0.000	0.000	-0.001
3.000	3.001	2.999
6.000	6.003	5.998
9.000	9.004	8.998
12.000	12.006	11.998
15.000	15.006	14.997

Applied (psi)	Reading # 11 (psi)	# 12 (psi)	# 13 (psi)	# 14 (psi)	# 15 (psi)	# 16 (psi)
0.000	0.000	0.000	-0.003	0.000	0.002	-0.004
20.000	20.000	19.990	19.991	19.995	19.993	19.992
40.000	40.001	39.989	39.984	39.988	39.995	39.987
60.000	59.986	59.984	59.968	59.976	59.993	59.975
80.000	79.989	79.974	79.964	79.963	79.985	79.953
100.000	99.961	99.961	99.956	99.950	99.972	99.944
60.000	59.997	59.992	59.987	59.987	60.000	59.981
0.000	0.014	0.011	0.013	0.016	0.016	0.016



Assignment

2019-03-08

Michael Karlsson

Responsible for Calibration

SIEMENS



CALIBRATION CERTIFICATE

issued by an Accredited Calibration Laboratory
KALIBRERINGSBEVIS utfärdad av Akkrediterat Kalibreringslaboratorium

Number 1853-193047	Sidan 1 av 2
Date of Calibration 2019-03-08	

Unit Under Test: Rosemount 3051 CA1 Abs. Pressure Transmitter
Asset Number: TC114046 , Label: p0
Serial Number: 1017935
Used Software: Fluke Met/Cal Version 7.20
Used Procedure : Transmitter: 80-120 kPa(a) CAL VER PPC4
Procedure Revision: 1.2
Warm-up time: > 1h (in laboratory environment).

Test Result: ¹⁾ **PASS**
Data Type: AS-LEFT
Ambient temperature: 23 ± 2 °C
Relative Humidity: 29 ± 10 %
Pressure Medium: Nitrogen Gas
Local g: 9,817939 m/s2
Ambient Atm. Pressure: hPa

Customer Siemens Industrial Turbomachinery AB, SE-612 44 Finspong.

Traceability The laboratory reference standards for electrical and pressure quantities are through regularly performed calibrations directly traceable to the Research Institutes of Sweden AB, RISE in Borås.

Method of calibration The object was calibrated with the standards at ambient atmosph. pressure. The supply voltage was +24VDC and the output current was measured as a voltage drop across the 100 ohm precision resistor. The object was calibrated acc. to procedure ICSI11876.

Calibration Uncertainty The reported expanded uncertainty of measurement is stated as the standard uncertainty of measurement multiplied by the coverage factor k=2 which for a normal distribution corresponds to a coverage probability of approximately 95%. The standard uncertainty has been determined in accordance with EAL Publication EA-4/02. The expanded uncertainty include uncertainty originating from the measurement standard by ±0,010% of applied pressure + 70 Pa and from the electrical measurement by ±0,005% of applied pressure.

Result This result is valid at the date of calibration. Measurements of long time instability is not performed. The Indicated Voltage is a mean value of at least three readings. The Nonlinearity and Hysteresis are calculated as percent of full scale.
¹⁾ Test Result "PASS" indicates that the object complies with the specification at the measured points.

Specifications used Siemens transmitters: SITRANS P Operating Instructions 09/12, A5E00047092-08.
 Rosemount transmitters: Rosemount 3051 Product Data Sheet 00813-0100-4001, Rev TC.

Remarks:

Standards Used			
Asset #	Description	Cal Date	Due Date
TC113248	Cropico RS3-100 Precision Resistor 100 ohms	2018-05-18	2019-05-17
TC119076	Druck DPI142 Precision Barometer	2018-09-05	2019-09-05
TC120344	Agilent 34401A 6,5 Digit Multimeter	2018-05-17	2019-05-17
TC123351	DHI PPC4 Pressure Controller	2018-10-03	2019-10-03
TC127301	Siemens 4-channel Transmitter Calibration Unit	2018-05-17	2019-05-17

Laboratories are accredited by the Swedish Board for Accreditation and Conformity Assessment (SWEDAC) under the terms of Swedish legislation. The accredited laboratory activities meet the requirements in SS-EN ISO/IEC 17025 (2005)

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 Number
 1853-193047

Sidan 2 av 2

Test Results

Applied Pressure	Indicated Voltage	Lower limit	Upper limit	Expanded Uncertainty [kPa]	Pass/Fail
80 kPa	0.4003	0.3980	0.4020 V	0.045	Pass
84 kPa	0.5599	0.5572	0.5628 V	0.046	Pass
88 kPa	0.7201	0.7164	0.7236 V	0.046	Pass
92 kPa	0.8801	0.8756	0.8844 V	0.046	Pass
96 kPa	1.0398	1.0348	1.0452 V	0.046	Pass
100 kPa	1.2002	1.1940	1.2060 V	0.047	Pass
104 kPa	1.3597	1.3532	1.3668 V	0.047	Pass
108 kPa	1.5201	1.5124	1.5276 V	0.047	Pass
112 kPa	1.6803	1.6716	1.6884 V	0.047	Pass
116 kPa	1.8399	1.8308	1.8492 V	0.047	Pass
120 kPa	2.0003	1.9900	2.0100 V	0.048	Pass
116 kPa	1.8399	1.8308	1.8492 V	0.047	Pass
112 kPa	1.6801	1.6716	1.6884 V	0.047	Pass
108 kPa	1.5199	1.5124	1.5276 V	0.047	Pass
104 kPa	1.3598	1.3532	1.3668 V	0.047	Pass
100 kPa	1.1999	1.1940	1.2060 V	0.047	Pass
96 kPa	1.0398	1.0348	1.0452 V	0.046	Pass
92 kPa	0.8798	0.8756	0.8844 V	0.046	Pass
88 kPa	0.7198	0.7164	0.7236 V	0.046	Pass
84 kPa	0.5598	0.5572	0.5628 V	0.046	Pass
80 kPa	0.3999	0.3980	0.4020 V	0.045	Pass
Nonlinearity					
0.0000 %	0.0094	-0.1000	0.1000 %		Pass
Hysteresis					
0.0000 %	0.0072	-0.1000	0.1000 %		Pass
Polynom					
BSL: $P(\text{kPa}) = U(\text{V}) * 24.9990758314 + 69.9994085197$					

The object is labeled with recommended date for recalibration.

Assignment: 2019-03-08

Filip Jacobson

Respons. for Calibration

Anders Liljeberg

Respons. for Laboratory



Akkred.nr 0076
Kalibrering
ISO/IEC 17025

KALIBRERINGSBEVIS

utfärdat av ackrediterat kalibreringslaboratorium
CALIBRATION CERTIFICATE issued by an Accredited Calibration Laboratory

Kalibreringsdatum:
2019-05-08

Bevis nummer:
0076-19334

Pentronic ordernr:
174536



Sida(sidantal):
1(2)

Uppdragsgivare:
Siemens Industrial Turbomachinery AB
Östermalmsvägen 1
612 40 FINSPÅNG

Kundreferens:
7000869443

Kontaktperson:

Kalibrering av 9 st resistanstermometrar typ Pt-100

Identifiering:

Objekt: Resistanstermometer typ Pt-100, Pentronic s/n 19191039 - 19191047.

Ankomstkick:

Objekt utan synliga skador.

Kalibreringsplats:

Akkrediterat kalibreringslaboratorium nr 0076, Pentronic AB, Verkeback.

Metod:

Jämförelsekalibrering i temperaturstabila bad/ugnar enligt vår metod 2.60.
Kalibreringen har utförts i fallande temperaturföljd vid nedanstående nivåer.

Mätmiljö mm:

Temperaturskala: ITS-90.
Omgivningstemperatur: $23\text{ }^{\circ}\text{C} \pm 2\text{ }^{\circ}\text{C}$.
Mätström: 1 mA.

Resultat:

I tabellen angivna värden är vid varje mättemperatur grundade på medelvärden av minst 10 observationer, med ett intervall av ca 10 sekunder.

Anvisning för korrektionstermen: Korrigerat värde = avläst värde + korrektion.

Resistans för nedanstående Pt-100 vid kalibreringstemperaturerna:

Pentronic s/n	45 °C		Utvidgad mätosäkerhet $\pm\text{ }^{\circ}\text{C}$
	Resistans ohm	Korrektion °C	
19191039	117,4800	-0,025	0,015
19191040	117,4815	-0,029	0,015
19191041	117,4996	-0,076	0,015
19191042	117,4971	-0,069	0,015
19191043	117,4949	-0,064	0,015
19191044	117,4836	-0,034	0,015
19191045	117,4851	-0,038	0,015
19191046	117,4981	-0,072	0,015
19191047	117,4760	-0,015	0,015

Postadress
Pentronic AB
593 96 Västervik

7th/Fax
0490-25 85 20
0490-237 66

E-post/Internet
aki@pentronic.se
www.pentronic.se

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of the issuing laboratory.

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KALIBRERINGSBEVIS
CALIBRATION CERTIFICATE

Kalibreringsdatum:
2019-05-08

Bevis nummer:
0076-19334



Mätosäkerhet:

Den angivna utvidgade mätosäkerheten är produkten av standardmätosäkerheten och täckningsfaktorn $k=2$, vilket för en normalfördelning svarar mot en täckningssannolikhet av ungefär 95 %. I standardosäkerheten ingår uppskattade osäkerhetsbidrag från alla faktorer som ansetts påverka mätningen. Standardmätosäkerheten har bestämts i enlighet med EA:s publikation EA-4/02. Hänsyn har ej tagits till det kalibrerade objektets långtidsstabilitet.

Spårbarhet:

Laboratoriets referensnormaler för temperatur och elektriska storheter är direkt spårbara till Riksmätplatsen för temperatur och Riksmätplatsen för elektriska storheter, båda hos SP i Borås.

Använd utrustning:

Resistansmätbrygga Isotech MicroK 400 s/n 07-P013 ITL 27888/1
Referenstermometer Isotech typ 670SQ s/n 191
Vattenbad HETO typ KB21 s/n 92121631

Utfört av


Karoline Harneck

Granskat av


Andreas Holm

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